

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
Electronics and Communication Engineering
Basic Electronics (24B11EC111)
Tutorial No. 1

- Q1. Using only $1\text{ k}\Omega$ resistors, synthesize a resistor of $3/5\text{ k}\Omega$ and $5/3\text{ k}\Omega$. You can use maximum 4 resistors in each case.
- Q2. Find the effective resistance between terminals A and B for the networks given in Fig. 1.

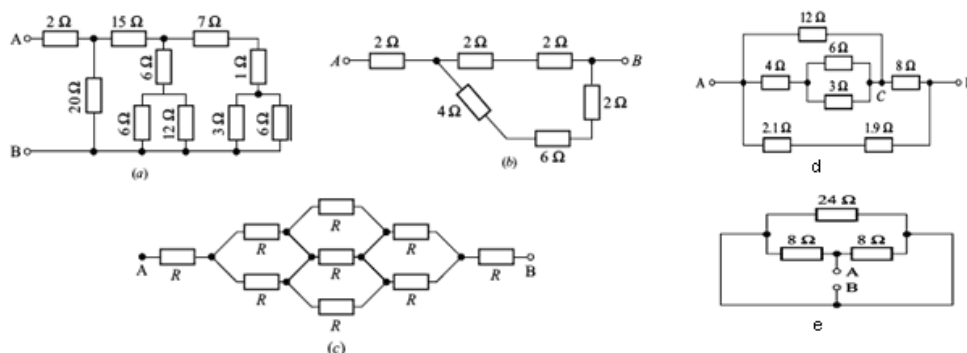


Figure 1

[Ans. (a) $12\text{ }\Omega$; (b) $5\text{ }\Omega$; (c) $10R/3$ (d) $3\text{ }\Omega$ (e) $4\text{ }\Omega$]

- Q3. Calculate the current drawn from a 12-V supply with internal resistance $0.5\text{ }\Omega$ by the infinite ladder network, each resistance being 1 ohm , in Fig. 2.

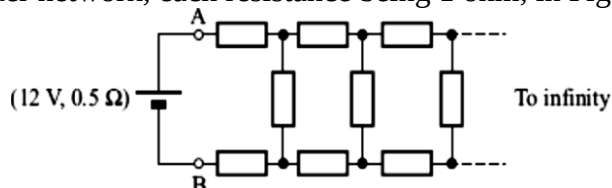


Figure 2

[Ans. 3.71 A]

- Q4. Using Kirchhoff's laws, determine the values of i_2 , i_4 , v_2 , v_3 , and v_6 in Fig. 3.

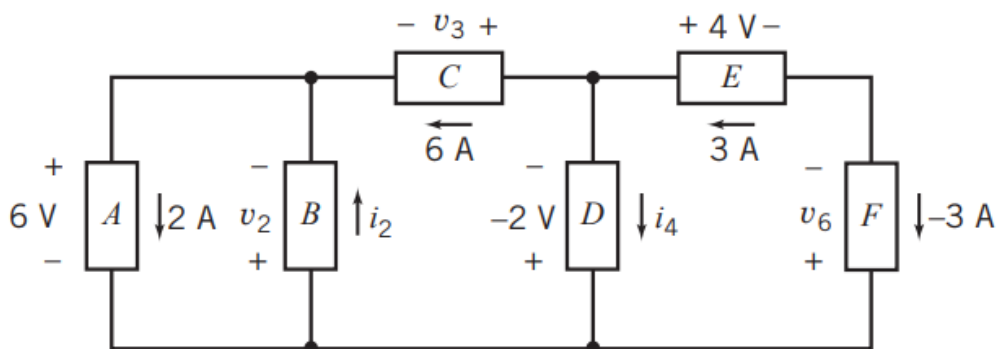


Figure 3

[Ans. $i_2 = -4\text{ A}$, $i_4 = -3\text{ A}$, $v_2 = -6\text{ V}$, $v_4 = -4\text{ V}$, $v_6 = 2\text{ V}$]

- Q5. Consider the two similar voltage divider circuits shown in Fig. 4. Use voltage division to determine the values of the voltage v_2 in Fig. 4(a) and the voltage v_b in Fig. 4(b).

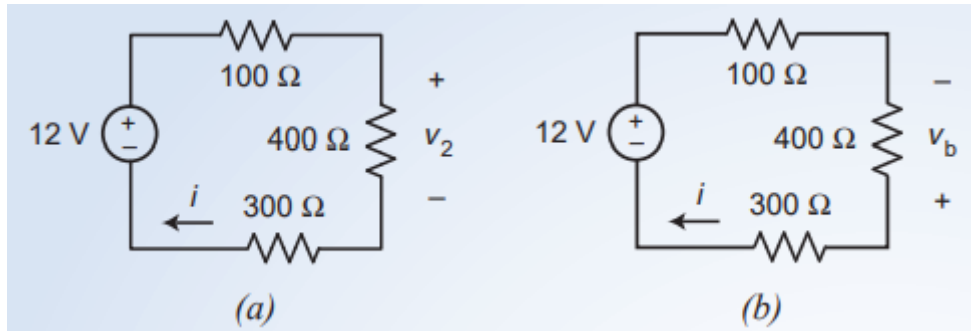


Figure 4

[Ans. $v_2 = 6\text{V}$ and $v_b = -6\text{V}$]

- Q6. Use current division to determine the currents i_1 , i_2 , i_3 , and i_4 in the circuit shown in Fig. 5

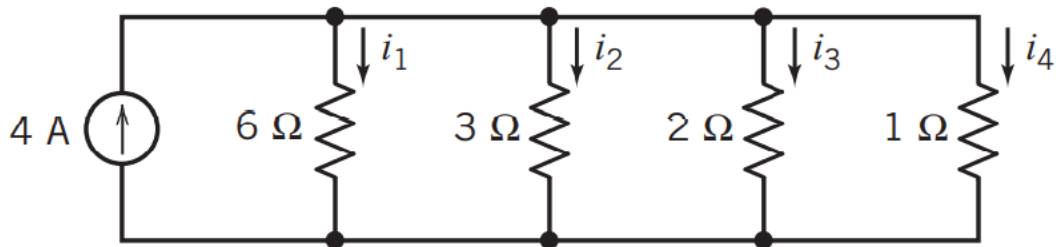


Figure 5

[Ans. $i_1 = 1/3\text{ A}$, $i_2 = 2/3\text{ A}$, $i_3 = 1\text{ A}$, $i_4 = 2\text{ A}$]

- Q7. Determine the node voltage at nodes 'a' and 'b' in the network shown in Fig. 6.

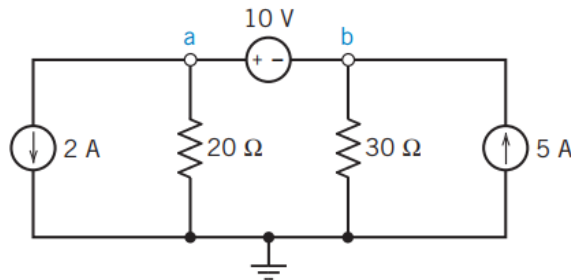


Figure 6

[Ans: $V_a = 40\text{ V}$, $V_b = 30\text{ V}$]

- Q8. Consider the circuit of Fig. 7 where $R_1 = R_2 = 1\ \Omega$ and $R_3 = 2\ \Omega$. Find the three mesh currents.

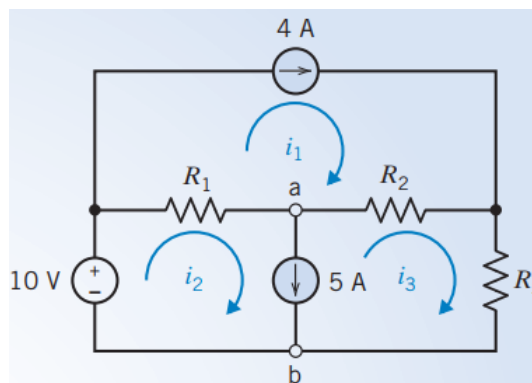


Figure 7

[Ans: $i_1 = 4\text{ A}$, $i_2 = 33/4\text{ A}$, $i_3 = 13/4\text{ A}$]

Basic Electronics

Tutorial-1 Solution

Solution 1

3/5 K and 5/3 K



Solution 2

(a) The resistance of the last branch is $R_4 = 8 + (3 \parallel 6) = 10 \Omega$. The resistance of the second branch is $R_3 = 6 + (6 \parallel 12) = 10 \Omega$. These two are connected in parallel, to give a resistance of $10/2 = 5 \Omega$. Now, $(15 + 5) \parallel 20 = 10 \Omega$. Therefore, $R_{AB} = 2 + 10 = 12 \Omega$.

(b) $R_{AB} = 2 + [(2 + 2) \parallel (4 + 6 + 2)] = 2 + [4 \parallel 12] = 5 \Omega$.

(c) $R_{AB} = R + R/2 + R/3 + R/2 + R = 10R/3$.

(d) $R_{AC} = 12 \parallel [4 + (6 \parallel 3)] = 4 \Omega$; $R_{AB} = (4 + 8) \parallel (2.1 + 1.9) = 3 \Omega$

(e) The 24-ohm resistance is shorted. Thus, 8-ohm and 8-ohm resistances are in parallel.

Solution 3

Removing the first three resistances of the infinite ladder, the remaining ladder network has an equivalent resistance equal to that of the original ladder. Removing one rung of an infinite ladder does not change its resistance R_x . Hence,

$$R_x = 1 + (1 \parallel R_x) + 1 \quad \text{or} \quad R_x = 2 + \frac{1 \times R_x}{1 + R_x} \quad \text{or} \quad R_x^2 - 2R_x - 2 = 0 \quad R_x = 2.73 \Omega; \quad I = \frac{12}{0.5 + 2.73} = 3.71 \text{ A}$$

Solution 4

Apply KCL at node a to get $2 = i_2 + 6 = 0 \Rightarrow i_2 = -4 \text{ A}$

Apply KCL at node b to get $3 = i_4 + 6 \Rightarrow i_4 = -3 \text{ A}$

Apply KVL to the loop consisting of elements A and B to get

$$-v_2 - 6 = 0 \Rightarrow v_2 = -6 \text{ V}$$

Apply KVL to the loop consisting of elements C , D , and A to get

$$-v_3 - (-2) - 6 = 0 \Rightarrow v_4 = -4 \text{ V}$$

Apply KVL to the loop consisting of elements E , F and D to get

$$4 - v_6 + (-2) = 0 \Rightarrow v_6 = 2 \text{ V}$$

Check: The sum of the power supplied by all branches is

$$-(6)(2) - (-6)(-4) - (-4)(6) + (-2)(-3) + (4)(3) + (2)(-3) = -12 - 24 + 24 + 6 + 12 - 6 = 0$$

Solution 5

- First, consider the circuit shown in Figure 1.5a. This circuit is an example of a single loop circuit like the circuit shown in Figure 1.5. The 100 , 400 , and 300Ω resistors are connected in series. The current in the loop is given by

$$i = \frac{12}{100+400+300} = 0.015 \text{ A} = 15\text{mA}$$

- We can calculate the value of v_2 using voltage division:

$$v_2 = \frac{400}{100+400+300}(12) = 6\text{V}$$

- As a check, notice that $6 = v_2 = 400(i) = 400(0.015)$
- Next, consider the circuit shown in Figure 1.5b. This circuit is also an example of a single loop circuit. Again, the current in the loop is given by

$$i = \frac{12}{100 + 400 + 300} = 0.015 \text{ A} = 15\text{mA}$$

- Notice that the voltage v_b in Figure 1.5b is the same voltage as the voltage v_2 in Figure 1.5a, except for polarity. Consequently

$$v_2 = -v_b$$

- Therefore

$$v_b = \frac{400}{100 + 400 + 300}(12) = -6\text{V}$$

Solution 6

$$i_1 = \frac{\frac{1}{6}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} 4 = \frac{1}{1+2+3+6} 4 = \underline{\underline{\frac{1}{3} \text{ A}}}$$

$$i_2 = \frac{\frac{1}{3}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} 4 = \underline{\underline{\frac{2}{3} \text{ A}}};$$

$$i_3 = \frac{\frac{1}{2}}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} 4 = \underline{\underline{1 \text{ A}}}$$

$$i_4 = \frac{1}{\frac{1}{6} + \frac{1}{3} + \frac{1}{2} + \frac{1}{1}} 4 = \underline{\underline{2 \text{ A}}}$$

Solution 7

$$2 + \frac{v_b + 10}{20} + \frac{v_b}{30} = 5 \Rightarrow v_b = 30 \text{ V} \text{ and } v_a = 40 \text{ V}$$

Solution 8

Because the 4-A source is in mesh 1 only, we note that

$$i_1 = 4$$

For the 5-A source, we have

$$i_2 - i_3 = 5 \quad (4.6-6)$$

Writing KVL for mesh 2 and mesh 3, we obtain

$$\text{mesh 2: } R_1(i_2 - i_1) + v_{ab} = 10 \quad (4.6-7)$$

$$\text{mesh 3: } R_2(i_3 - i_1) + R_3 i_3 - v_{ab} = 0 \quad (4.6-8)$$

We substitute $i_1 = 4$ and add Eqs. 4.6-7 and 4.6-8 to obtain

$$R_1(i_2 - 4) + R_2(i_3 - 4) + R_3 i_3 = 10 \quad (4.6-9)$$

From Eq. 4.6-6, $i_2 = 5 + i_3$, substituting into Eq. 4.6-9, we have

$$R_1(5 + i_3 - 4) + R_2(i_3 - 4) + R_3 i_3 = 10$$

Using the values for the resistors, we obtain

$$i_3 = \frac{13}{4} \text{ A and } i_2 = 5 + i_3 = \frac{33}{4} \text{ A}$$

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Tutorial No. 2

Q1. Using the superposition theorem, find in the circuit of Fig. 1.

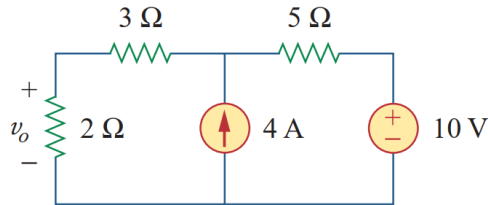


Figure 1

[Ans. 6 V]

Q2. For the circuit in Fig. 2, use the superposition theorem to find i .

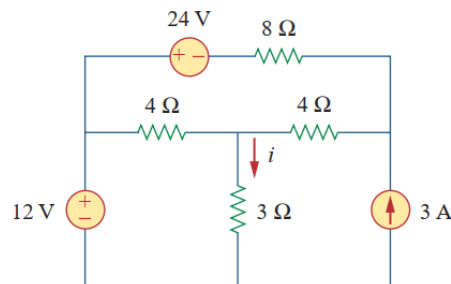


Figure 2

[Ans. 2 A]

Q3. Find the Thevenin equivalent circuit of the circuit shown in Fig. 3, to the left of the terminals a-b. Then find the current through $R_L=6\ \Omega$.

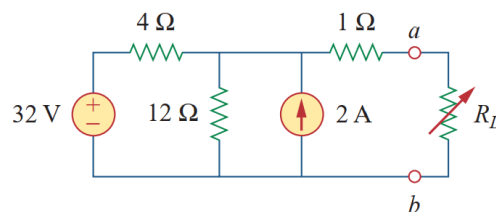


Figure 3

[Ans. $R_{TH}=4\ \Omega$, $V_{TH}=30\text{ V}$, $I_L=3\text{ A}$]

Q4. Find the Thevenin equivalent of the circuit in Fig. 4 at terminals a-b.

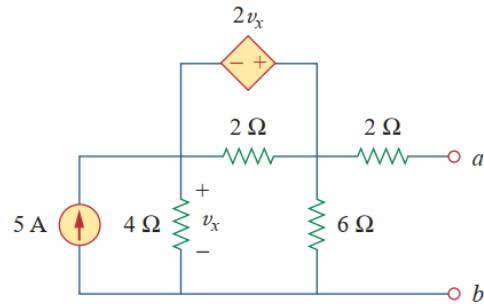


Figure 4

[Ans. $R_{TH} = 6 \Omega$, $V_{TH} = 20 \text{ V}$]

Q5. Find the Norton equivalent circuit across terminal a-b in following Fig. 5.

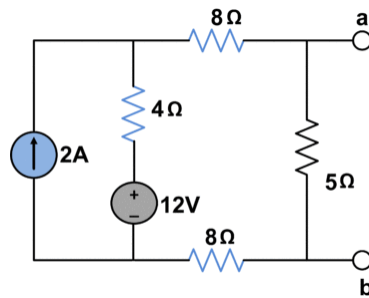


Figure 5

[Ans. $R_N = 4 \Omega$, $I_N = 1 \text{ A}$]

Q6. For the circuit shown in Fig. 6, find the load current using Norton theorem.

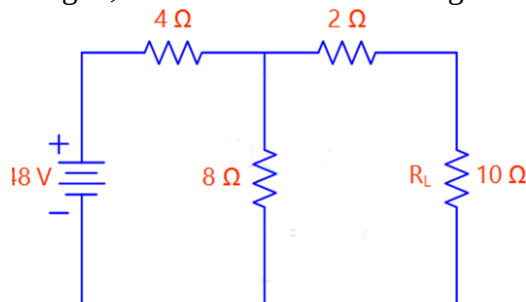


Figure 6

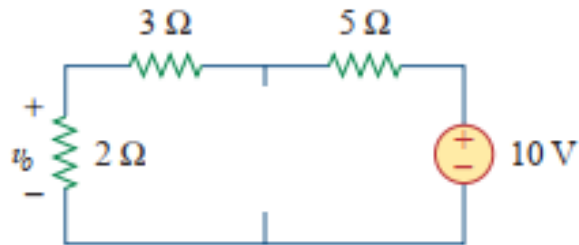
[Ans. $I_L = 2.18 \text{ A}$]

Basic Electronics

Tutorial-2 Solution

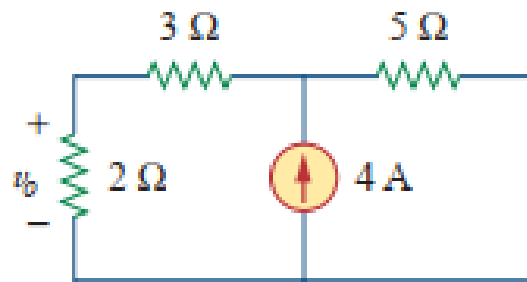
Solution 1

Case 1: Consider 10 V source only



$$V_{b1} = \frac{2}{2+3+5} \times 10 = 2V$$

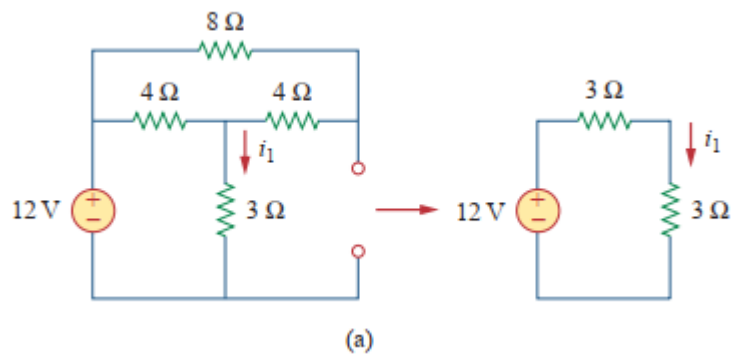
Case 2: Consider 4 A source only

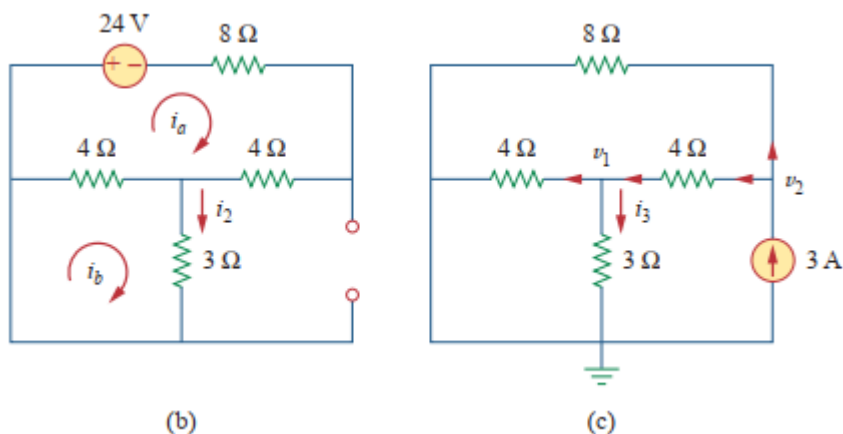


$$V_{b2} = 2 \times \frac{5}{2+3+5} \times 4 = 4V$$

$$v_b = v_{b1} + v_{b2} = 6V$$

Solution 2





In this case, we have three sources. Let

$$i = i_1 + i_2 + i_3$$

where i_1 , i_2 , and i_3 are due to the 12-V, 24-V, and 3-A sources respectively. To get i_1 , consider the circuit in Fig. 4.13(a). Combining $4\ \Omega$ (on the right-hand side) in series with $8\ \Omega$ gives $12\ \Omega$. The $12\ \Omega$ in parallel with $4\ \Omega$ gives $12 \times 4/16 = 3\ \Omega$. Thus,

$$i_1 = \frac{12}{6} = 2\text{ A}$$

To get i_2 , consider the circuit in Fig. 4.13(b). Applying mesh analysis gives

$$16i_a - 4i_b + 24 = 0 \quad \Rightarrow \quad 4i_a - i_b = -6 \quad (4.5.1)$$

$$7i_b - 4i_a = 0 \quad \Rightarrow \quad i_a = \frac{7}{4}i_b \quad (4.5.2)$$

Substituting Eq. (4.5.2) into Eq. (4.5.1) gives

$$i_2 = i_b = -1$$

To get i_3 , consider the circuit in Fig. 4.13(c). Using nodal analysis gives

$$3 = \frac{v_2}{8} + \frac{v_2 - v_1}{4} \quad \Rightarrow \quad 24 = 3v_2 - 2v_1 \quad (4.5.3)$$

$$\frac{v_2 - v_1}{4} = \frac{v_1}{4} + \frac{v_1}{3} \quad \Rightarrow \quad v_2 = \frac{10}{3}v_1 \quad (4.5.4)$$

Substituting Eq. (4.5.4) into Eq. (4.5.3) leads to $v_1 = 3$ and

$$i_3 = \frac{v_1}{3} = 1\text{ A}$$

Thus,

$$i = i_1 + i_2 + i_3 = 2 - 1 + 1 = 2\text{ A}$$

Solution 3

We find by turning off the 32-V voltage source (replacing it with a short circuit) and the 2-A current source (replacing it with an

open circuit). The circuit becomes what is shown in Fig. 4.28(a). Thus,

$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4 \Omega$$

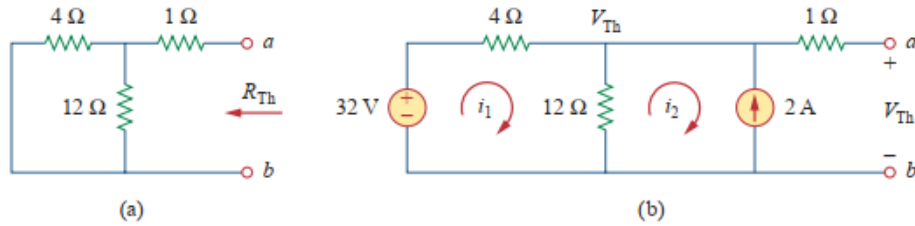


Figure 4.28

For Example 4.8: (a) finding R_{Th} , (b) finding V_{Th} .

To find V_{Th} , consider the circuit in Fig. 4.28(b). Applying mesh analysis to the two loops, we obtain

$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

Solving for i_1 , we get $i_1 = 0.5 \text{ A}$. Thus,

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$

Alternatively, it is even easier to use nodal analysis. We ignore the 1-Ω resistor since no current flows through it. At the top node, KCL gives

$$\frac{32 - V_{Th}}{4} + 2 = \frac{V_{Th}}{12}$$

or

$$96 - 3V_{Th} + 24 = V_{Th} \Rightarrow V_{Th} = 30 \text{ V}$$

as obtained before. We could also use source transformation to find V_{Th} .

The Thevenin equivalent circuit is shown in Fig. 4.29. The current through R_L is

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

When $R_L = 6$,

$$I_L = \frac{30}{10} = 3 \text{ A}$$

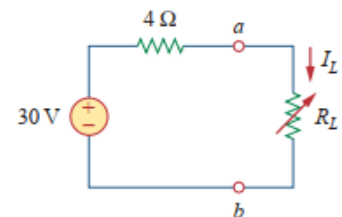


Figure 4.29

The Thevenin equivalent circuit for Example 4.8.

Solution 4

This circuit contains a dependent source, unlike the circuit in the previous example. To find R_{Th} , we set the independent source equal to zero but leave the dependent source alone. Because of the presence of the dependent source, however, we excite the network with a voltage source v_o connected to the terminals as indicated in Fig. 4.32(a). We may set $v_o = 1$ V to ease calculation, since the circuit is linear. Our goal is to find the current i_o through the terminals, and then obtain $R_{Th} = 1/i_o$. (Alternatively, we may insert a 1-A current source, find the corresponding voltage v_o , and obtain $R_{Th} = v_o/1$.)

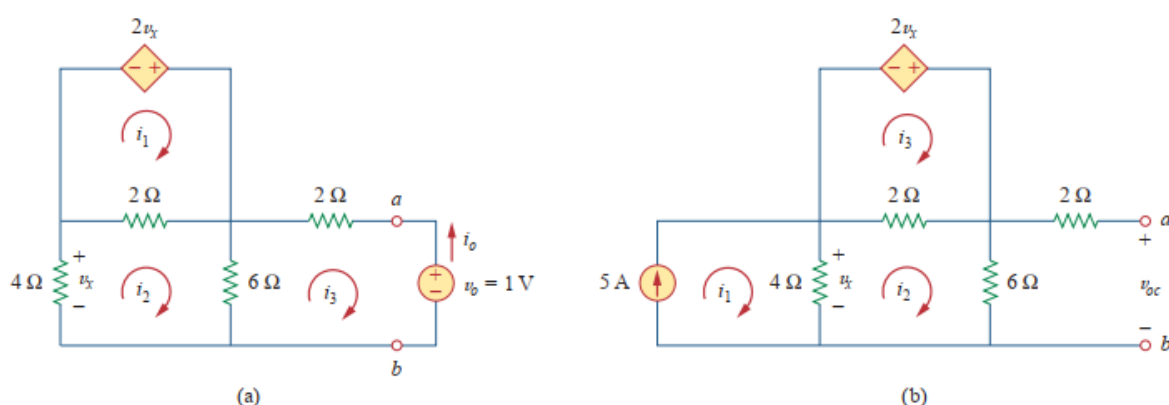


Figure 4.32
Finding R_{Th} and V_{Th} for Example 4.9.

Applying mesh analysis to loop 1 in the circuit of Fig. 4.32(a) results in

$$-2v_x + 2(i_1 - i_2) = 0 \quad \text{or} \quad v_x = i_1 - i_2$$

But $-4i_2 = v_x = i_1 - i_2$; hence,

$$i_1 = -3i_2 \quad (4.9.1)$$

For loops 2 and 3, applying KVL produces

$$4i_2 + 2(i_2 - i_1) + 6(i_2 - i_3) = 0 \quad (4.9.2)$$

$$6(i_3 - i_2) + 2i_3 + 1 = 0 \quad (4.9.3)$$

Solving these equations gives

$$i_3 = -\frac{1}{6} \text{ A}$$

But $i_o = -i_3 = 1/6 \text{ A}$. Hence,

$$R_{\text{Th}} = \frac{1 \text{ V}}{i_o} = 6 \Omega$$

To get V_{Th} , we find v_{oc} in the circuit of Fig. 4.32(b). Applying mesh analysis, we get

$$i_1 = 5 \quad (4.9.4)$$

$$-2v_x + 2(i_3 - i_2) = 0 \quad \Rightarrow \quad v_x = i_3 - i_2 \quad (4.9.5)$$

$$4(i_2 - i_1) + 2(i_2 - i_3) + 6i_2 = 0$$

or

$$12i_2 - 4i_1 - 2i_3 = 0 \quad (4.9.6)$$

But $4(i_1 - i_2) = v_x$. Solving these equations leads to $i_2 = 10/3$. Hence,

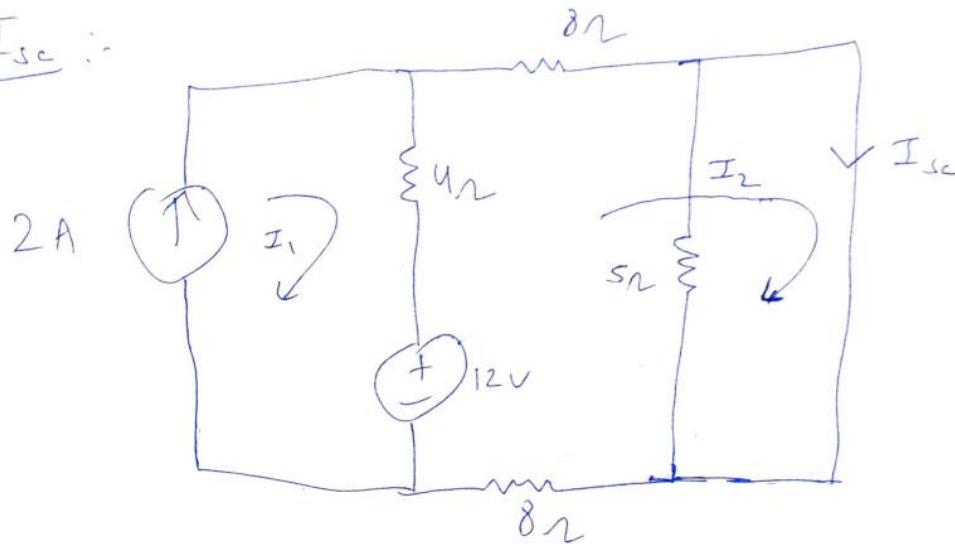
$$V_{\text{Th}} = v_{oc} = 6i_2 = 20 \text{ V}$$

The Thevenin equivalent is as shown in Fig. 4.33.

Solution 5

$$R_N = 5 \parallel (8 + 4 + 8) \\ = 4 \Omega$$

I_{sc} :

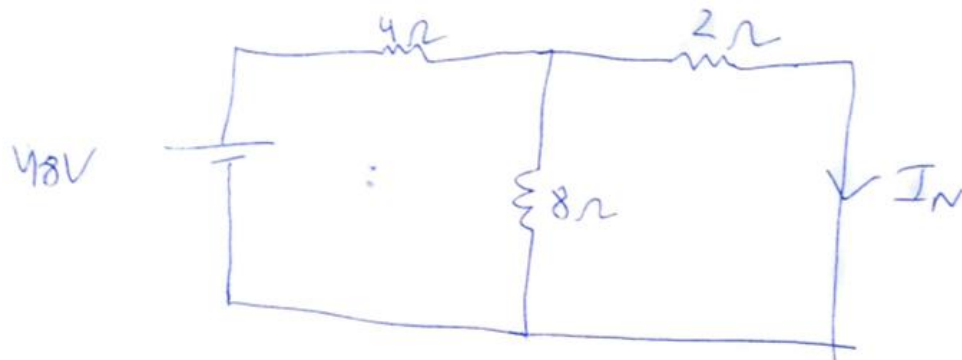


$$\boxed{i_1 = 2A}$$

$$20 i_2 - 4 i_1 - 12 = 0$$

$$\Rightarrow \overline{i_1 = 1A}$$

Solution 6

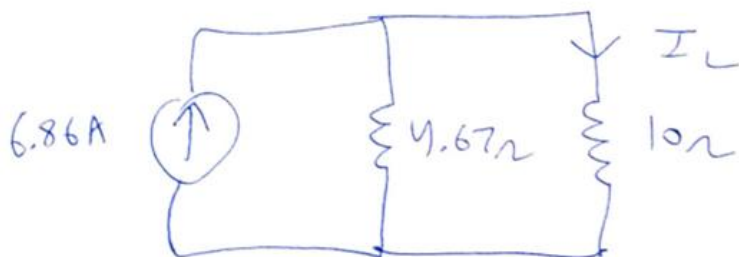


$$R_{eq} = 4 + \frac{8 \times 2}{8 + 2} = 5.6 \Omega$$

$$I_t = \frac{48}{5.6} = 8.57 \text{ A}$$

$$I_N = 8.57 \times \frac{8}{8 + 2} = 6.86 \text{ A}$$

$$R_N = \frac{4 \times 8}{4 + 8} + 2 = 4.67 \Omega$$



$$I_L = 6.86 \times \frac{4.67}{4.67 + 10}$$

$$I_L = 2.18 \text{ A}$$

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Tutorial No. 3

Q1. The current flowing in a PN-junction diode at room temperature is 200 nA, when a large reverse bias is applied. Calculate the current through the diode when 0.1 V forward bias is applied. [Ans. 9.1625 μ A]
[CO2]

Q2. In the circuit given in Fig. 1, a silicon diode with knee voltage 0.7 V is used. The dynamic resistance of the diode in forward bias may assumed to be zero.
[CO2]

- a) If $V_{DD} = 5$ V, what should be the value of R to establish a current of 5 mA in the circuit?
- b) Determine the power dissipated in the resistance R and in the diode D, when a current of 5 mA flows in the circuit at $V_{DD} = 6$ V.

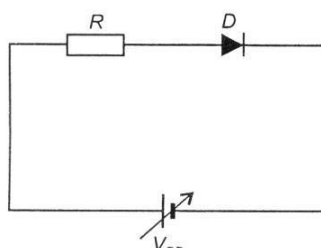


Figure 1

[Ans. (a) 860 Ω ; (b) 26.5 mW, 3.5 mW]

Q3. Germanium diodes D_1 and D_2 with knee voltage 0.3 V are used in the circuit given in Fig. 2. Determine the potential of point X with respect to point Y.
[CO2]

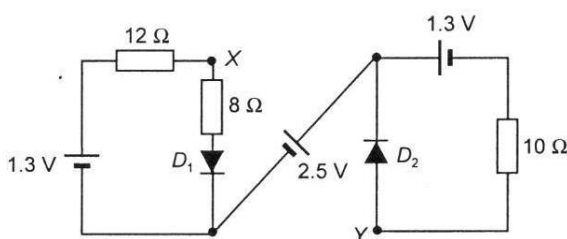


Figure 2

[Ans. -0.5 V]

Q4. The diodes used in the circuit Fig. 3 are identical with $V_T = 0.6$ V and $R_F = 10$ Ω .
[CO2] Determine V_D , I_D and V_o as marked in the figure.

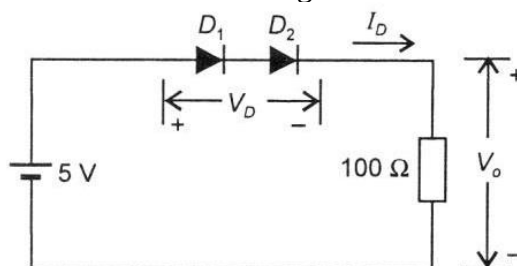


Figure 3

[Ans. 1.833 V, 31.667 mA, 3.167 V]

- Q5.** Determine the current through the $3\text{-}\Omega$ resistor in the circuit of Fig. 4. Assume the diode to be ideal.
[CO2]

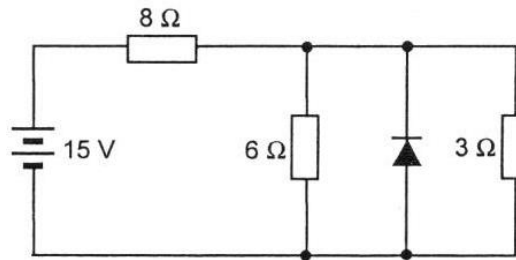


Figure 4

[Ans. 1 A]

- Q6.** If the terminals of the diode are reversed in the circuit of Fig. 4, what would be the current in the $3\text{-}\Omega$ resistor?
[CO2]

[Ans. zero]

- Q7.** Determine the current through the circuit of Fig. 5, assuming the diode to be a piecewise linear model with $V_T = 0.3\text{ V}$ and $r_f = 8\text{ }\Omega$.
[CO2]

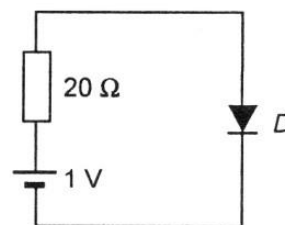


Figure 5

[Ans. 25 mA]

- Q8.** Determine the value of current through the $300\text{-}\Omega$ resistor in the circuit of Fig. 6. Assume that the diodes D_1 and D_2 are ideal.
[CO2]

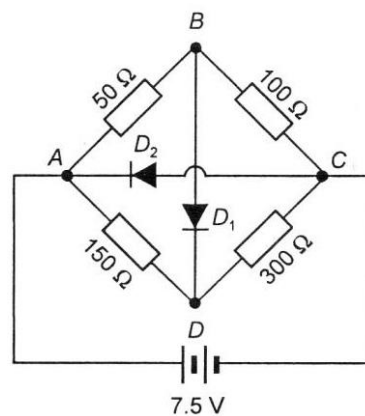


Figure 6

[Ans. 16.66 mA]

- Q9.** In the circuit given in Fig. 7, all diodes are identical and have a cut-in voltage (V_T) = 0.6 V.
[CO2]

- Find the voltage V_o and current I , for (i) $V_i = 0\text{ V}$, and (ii) $V_i = +2\text{ V}$.
- Plot the output voltage V_o , if the input voltage is $v_i = 2\sin 2\pi ft\text{ V}$, where $f = 50\text{ Hz}$

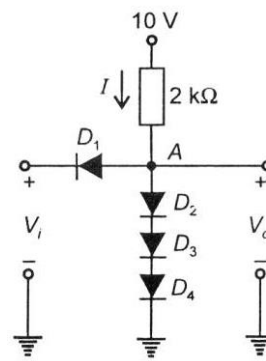


Figure 7

[Ans. (a) (i) 0.6 V, 4.7 mA; (ii) 1.8 V, 4.1 mA]

Solutions to Tutorial Sheet- 3

$$1. \quad I_D = I_s \left[\exp\left(\frac{V_D}{V_T}\right) - 1 \right] = 200 \times 10^{-9} \left[\exp\left(\frac{0.1}{0.026}\right) - 1 \right] = \mathbf{9.1625 \mu A}$$

$$2. (a) \quad V_{DD} = 5 \text{ V}, I = 5 \text{ mA}, V_T = 0.7 \text{ V}; V_R = V_{DD} - V_T = 5 - 0.7 = 4.3 \text{ V}$$

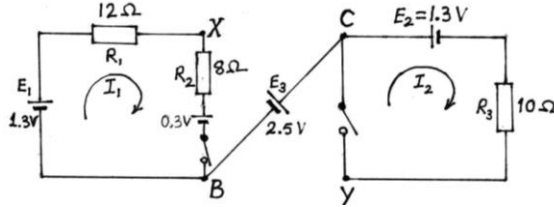
$$\therefore R = \frac{V_R}{I} = \frac{4.3}{5 \times 10^{-3}} = \mathbf{860 \Omega}$$

$$(b) \quad V_R = V_{DD} - V_T = 6 - 0.7 = 5.3 \text{ V}; \therefore R = \frac{5.3}{5 \times 10^{-3}} = 1060 \Omega$$

$$\text{Power dissipated in resistance } R = I^2 R = (5 \times 10^{-3})^2 \times 1060 = \mathbf{26.5 \text{ mW}}$$

$$\text{Power dissipated in the diode} = IV_T = 5 \times 10^{-3} \times 0.7 = \mathbf{3.5 \text{ mW}}$$

3. Equivalent circuit is shown in figure.



Since diode D1 is forward biased ($0.3 \text{ V} < 1.3 \text{ V}$), it is replaced by a closed switch and a battery of 0.3 V . Diode D2 is reverse-biased, it is replaced by an open switch. Therefore, $I_2 = 0$. Now,

$$I_1 = \frac{1.3 - 0.3}{8 + 12} = 0.05 \text{ A}; \quad V_{XB} = V_{XA} + V_{AB} = I_1 R_2 + V_{AB} = 0.05 \times 8 + 0.3 = 0.4 + 0.3 = 0.7 \text{ V}$$

$$\text{Now, } V_{CY} = I_2 R_3 + E_2 = 0 \times 10 + 1.3 = 1.3 \text{ V}; \therefore V_{XY} = V_{XB} + V_{BC} + V_{CY} = 0.7 - 2.5 + 1.3 = \mathbf{-0.5 \text{ V}}$$

4. Diodes are forward biased and hence each diode can be replaced by a resistance of 10Ω and a battery of 0.6 V .

$$I_D = \frac{5 - 0.6 - 0.6}{100 + 10 + 10} = \frac{5 - 1.2}{120} = \mathbf{31.667 \text{ mA}};$$

$$V_D = I_D (r_f + r_f) + V_T + V_T = 31.667 \times 10^{-3} (10 + 10) + 0.6 + 0.6 = \mathbf{1.833 \text{ V}}; \quad V_o = I_D \times 100 = \mathbf{3.167 \text{ V}}$$

5. Diode is reverse-biased, and hence it is an open-circuit. Hence,

$$\text{Net resistance} = \left[8 + \left(\frac{6 \times 3}{6 + 3} \right) \right] = 10 \Omega; \therefore I = \frac{15}{10} = 1.5 \text{ A}; \therefore I_1 = 1.5 \left(\frac{6}{6 + 3} \right) = \mathbf{1 \text{ A}}$$

6. The diode is now forward-biased and therefore it behaves as a short-circuit. Hence, the current through 3Ω resistor is **zero**.

$$7. \quad I = \frac{1 - 0.3}{20 + 8} = \frac{0.7}{28} = \mathbf{25 \text{ mA}}$$

8. Diode D2 is reversed biased, hence it will not conduct. Assume that the diode D1 is removed.

$$\therefore V_B = \left(\frac{100}{100 + 50} \right) 7.5 \text{ V} = 7.5 \left(\frac{2}{3} \right) \text{ V} \quad \text{and} \quad V_D = \left(\frac{300}{300 + 150} \right) 7.5 \text{ V} = 7.5 \left(\frac{2}{3} \right) \text{ V}$$

$$\text{Since } V_B \approx V_D, \text{ the diode } D_1 \text{ also does not conduct. } \therefore I_2 = \frac{7.5}{150 + 300} = \frac{7.5}{450} = \mathbf{16.66 \text{ mA}}$$

9. (a) When $V_i = 0 \text{ V}$, all diodes are likely to be forward-biased. Conduction through diode D1 brings the point A to 0.6 V . For the diodes D2, D3 and D4 to conduct, the minimum voltage required at point A is $0.6 \text{ V} + 0.6 \text{ V} + 0.6 \text{ V} = 1.8 \text{ V}$. Since $V_A = 0.6 \text{ V}$, these diodes do not conduct. So, these can be treated as open-circuit. Therefore, $V_o = \mathbf{0.6 \text{ V}}$.

$$\therefore I = \frac{10 - 0.6}{2 \times 10^3} = \mathbf{4.7 \text{ mA}}$$

(b) When $V_i = 2 \text{ V}$, the diodes D2, D3 and D4 conduct. This makes the voltage at point A as $V_A = 0.6 + 0.6 + 0.6 = 1.8 \text{ V}$. This makes the diode D1 reverse-biased and hence it cannot conduct.

$$\therefore V_o = \mathbf{1.8 \text{ V}} \quad \text{and} \quad I = \frac{10 - 1.8}{2 \times 10^3} = \frac{8.2}{2} = \mathbf{4.1 \text{ mA}}$$

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Basic Electronics (24B11EC111)
Tutorial No. 4

Q1. The turns ratio of a transformer used in a half-wave rectifier circuit is $N_1 : N_2 = 12:$
[CO2] 1. The primary is connected to the power mains (220 V, 50 Hz). Assuming the diode resistance in forward bias to be zero, calculate the dc voltage across the load. Also determine the PIV of the diode. **[Ans. 8.25 V, 25.93 V]**

Q2. A half-wave rectifier circuit uses a silicon diode with $r_f = 10 \Omega$ and $V_T = 0.7 \text{ V}$. The load
[CO2] connected at the output is $R_L = 500 \Omega$. The transformer has $N_1/N_2 = 10/1$. If the primary is connected to the ac mains supply (220 V, 50 Hz), calculate
 (a) the dc current through the load,
 (b) the rectification efficiency, and
 (c) the PIV rating of the diode.

[Ans. (a) 18.98 mA; (b) 39.8 %; (c) 31.1 V]

Q3. For the centre-tap rectifier circuit shown in Fig. 1, find
[CO2] (a) PIV rating of the diodes,
 (b) the maximum dc voltage that can be obtained, and
 (c) the reading of the milliammeter.

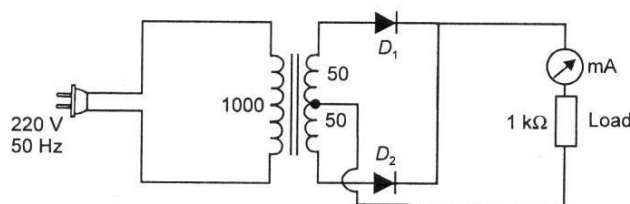


Figure 1

[Ans. (a) 31.1 V; (b) 9.9 V; (c) 9.9 mA]

Q4. The primary winding of the step-down transformer of a bridge rectifier is connected to ac
[CO2] mains (220 V, 50 Hz). It uses silicon diodes, with $V_T = 0.7 \text{ V}$. If it is desired to obtain a maximum dc voltage of 15 V from this circuit, find the turns ratio of the transformer. **[Ans. 12 : 1]**

Q5. The diodes used in a bridge rectifier circuit have the following parameters: $V_T = 0$
[CO2] V, $r_f = 10 \Omega$, $R_R = \infty$, $V_Z = \text{very large}$.
 The secondary winding of the transformer has a resistance $R_S = 10 \Omega$. A load of 970Ω is connected across its output. If the voltage across the secondary is $v_s = 20 \sin \omega t$ volts, determine
 (a) PIV rating of the diodes,
 (b) dc current through the load,
 (c) dc power supplied to the load, and
 (d) conversion efficiency of the circuit.

[Ans. (a) 19.6 V; (b) 12.7 mA; (c) 156.5 mW; (d) 79.56 %]

Q6. For the circuit of Fig. 2, sketch v_o and determine V_{dc} .
[CO2]

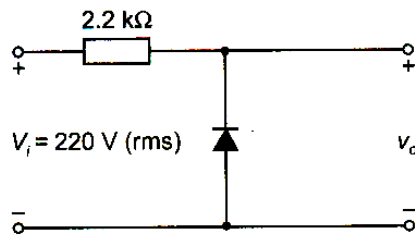


Figure 2

[Ans. 311 V (Peak), 99 V]

Q7. The circuit of Fig. 3 uses two silicon diodes in parallel. The maximum power rating of each diode, $P_{\max} = 14 \text{ mW}$. The input voltage v_i is a sinusoidal with peak value, $V_m = 150 \text{ V}$.

[CO2]

- Determine the maximum current rating of each diode.
- Determine $I_{\max}$ for $V_i(\max) = 150 \text{ V}$.
- Determine the current for each diode for $V_m = 150 \text{ V}$.
- Is the current determined in part (c) less than the current rating determined in part (a)?
- If only one diode were present, determine the diode current and compare it to the maximum current rating.

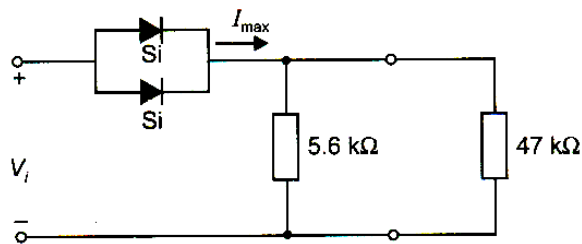


Figure 3

[Ans. (a) 20 mA; (b) 29.86 mA; (c) 14.93 mA; (d) Yes; (e) $I_D (=29.86 \text{ mA}) > I_{\max} (20 \text{ mA})$]

Q8. The input v_i to the circuit of Fig. 4 is a sinusoidal voltage of peak value 150 V. Assuming ideal diodes, sketch the output voltage v_o and determine its dc value.

[CO2]

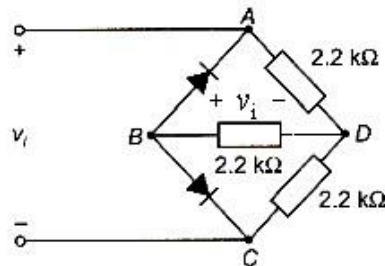


Figure 4

[Ans. Full-wave rectified wave with peak value = 50 V, -31.8 V]

Q9. The input v_i to the circuit of Fig. 5 is a floating sinusoidal voltage of peak-to-peak value 200V, obtained from an isolation transformer. Assuming ideal diodes, sketch the output voltage v_o and determine its dc value.

[CO2]

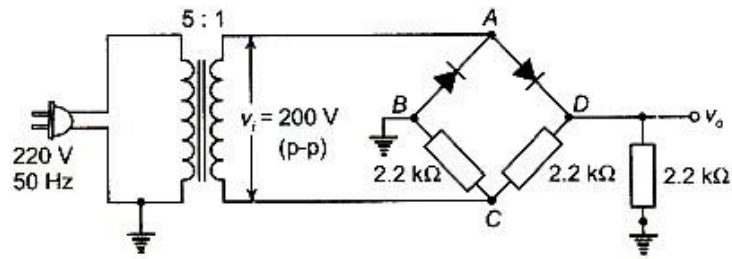


Figure 5

[**Ans.** Full-wave rectified wave with peak value = 50 V, 31.8 V]

Solutions to Tutorial Sheet-4

$$1. \quad V_2 = \frac{N_2}{N_1} \times V_1 = \frac{1}{12} \times 220 = 18.33 \text{ V}; \quad V_m = V_2 \times \sqrt{2} = 18.33 \times \sqrt{2} = 25.93 \text{ V}; \quad V_{dc} = \frac{V_m}{\pi} = \frac{25.92}{\pi} = 8.25 \text{ V};$$

$$PIV = V_m = 25.93 \text{ V}$$

$$2. (a) \quad V_2 = \frac{N_2}{N_1} V_1 = \frac{1}{10} \times 220 = 22 \text{ V}; \quad \therefore V_m = V_2 \times \sqrt{2} = 22 \times \sqrt{2} = 31.1 \text{ V}; \quad I_m = \frac{V_m - V_T}{R_L + r_f} = \frac{30.4}{510} \approx 59.6 \text{ mA};$$

$$\therefore I_{dc} = \frac{I_m}{\pi} = 18.98 \text{ mA}$$

$$(b) \quad \eta = \frac{40.6}{1 + r_f / R_L} \% = \frac{40.6}{1 + 10 / 500} \% = 39.8 \%$$

$$(c) \quad PIV = V_m - V_T = 31.1 - 0.7 = 31.1 \text{ V}$$

$$3. \quad N_1 : N_2 = 1000 : 50 = 20 : 1, \quad V_{in} = 220 \text{ V}, \quad R_L = 1 \text{ k}\Omega$$

$$V_2 = \frac{N_2}{N_1} V_1 = \frac{50}{1000} \times 220 = 11 \text{ V}; \quad V_m = V_2 \times \sqrt{2} = 15.5 \text{ V}$$

$$(a) \quad \therefore PIV = 2V_m = 31 \text{ V}; (b) \quad V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 31}{\pi} = 9.9 \text{ V}; (c) \quad I_{dc} = \frac{V_{dc}}{R_2} = \frac{9.9}{1000} = 9.9 \text{ mA}$$

$$4. \quad V_{dc} = \frac{2(V_m - 2V_T)}{\pi}; \quad \therefore V_m = \frac{\pi}{2} V_{dc} + 2V_T = \frac{\pi}{2} (15) + (0.7) = 24.95 \text{ V}; \quad V_2 = \frac{V_m}{\sqrt{2}} = 17.64 \text{ V};$$

$$\therefore \frac{N_1}{N_2} = \frac{V_1}{V_2} = \frac{220}{17.64} \approx 12.4; \quad \therefore N_1 : N_2 \approx 12:1$$

$$5. (a) \quad PIV = I_m (R_L + r_f) = 19.6 \text{ V}$$

$$(b) \quad I_m = (V_m - 2V_T) / (R_L + R_S + 2r_f) = \frac{20}{970 + 10 + 20} = 20 \text{ mA}; \quad I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 20}{\pi} \text{ mA} = 12.7 \text{ mA}$$

$$(c) \quad P_{dc} = I_{dc}^2 R_L = (12.7)^2 (970) \mu\text{W} = 156.5 \text{ mW}$$

$$(d) \quad \eta = \frac{81.2}{1 + 2r_f / R_L} \% = \frac{81.2}{1 + 20 / 970} \% = 79.56 \%$$

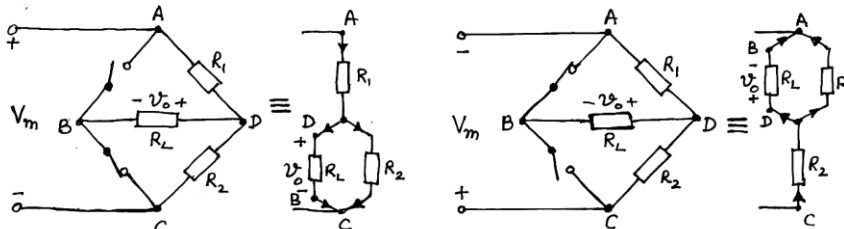
$$6. \quad V_m = 220 \times \sqrt{2} = 311 \text{ V}; \quad V_{dc} = \frac{V_m}{\pi} = 105.4 \text{ V}$$

$$7. (a) \quad I_{\max} = \frac{P_{\max}}{V_T} = \frac{14 \text{ mW}}{0.7 \text{ V}} = 20 \text{ mA}$$

$$(b) \quad R_L = 5.6 \text{ k}\Omega \parallel 47 \text{ k}\Omega = 5 \text{ k}\Omega; \quad I_{\max} = \frac{V_m - V_T}{R_L} = \frac{150 \text{ V} - 0.7 \text{ V}}{5 \text{ k}\Omega} = 29.86 \text{ mA}$$

$$(c) \quad I_D = \frac{I_{\max}}{2} = 14.93 \text{ mA}; (d) \text{ Yes}; (e) \quad I_D = 29.86 \text{ mA} > I_{\max} = 20 \text{ mA}$$

8.



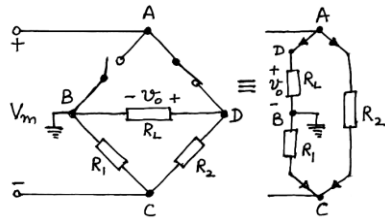
(a) For positive half-cycle.

(b) For negative half-cycle.

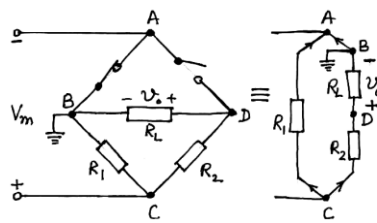
$$(a) \quad V_{o(\max)} = V_{DB(\max)} = V_m \frac{(R_L \parallel R_2)}{R_1 + (R_L \parallel R_2)} = (150 \text{ V}) \times \frac{1.1 \text{ k}\Omega}{2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega} = 50 \text{ V}$$

$$(b) \quad V_{o(\max)} = V_{DB(\max)} = V_m \frac{(R_L \parallel R_1)}{R_2 + (R_L \parallel R_1)} = (150 \text{ V}) \times \frac{1.1 \text{ k}\Omega}{2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega} = 50 \text{ V}; \text{ and } V_{dc} = \frac{2V_m}{\pi} = 31.8 \text{ V}$$

9.



(a) For positive half-cycle.



(b) For negative half-cycle.

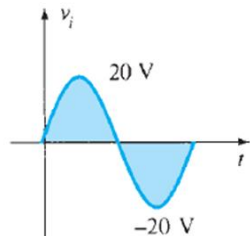
$$V_m = \frac{V_{pp}}{2} = \frac{200 \text{ V}}{2} = 100 \text{ V}$$

$$(a) \quad V_{o(\max)} = V_{DB(\max)} = V_m \frac{R_L}{R_L + R_1} = (100 \text{ V}) \times \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 50 \text{ V}$$

$$(b) \quad V_{o(\max)} = V_{DB(\max)} = V_m \frac{R_L}{R_L + R_2} = (100 \text{ V}) \times \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 50 \text{ V}; \text{ and } V_{dc} = \frac{2V_m}{\pi} = 31.8 \text{ V}$$

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Tutorial No. 5

- Q1.** Determine V_0 for each network as shown in Figure 1 and 2 for the input signal. Consider diode is a silicon diode.
[CO2]



Input Signal

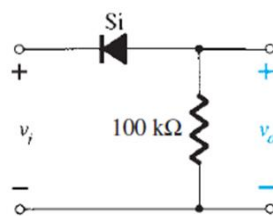


Figure 1

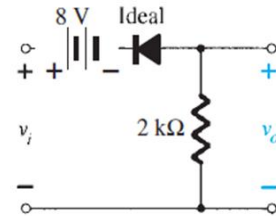
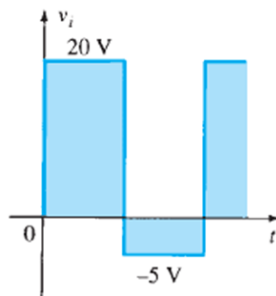


Figure 2

- Q2.** Determine V_0 for each network as shown in Figure 3 and 4 for the input signal. Consider diode is a silicon diode.
[CO2]



Input Signal

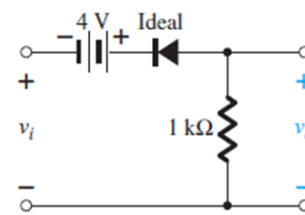


Figure 3

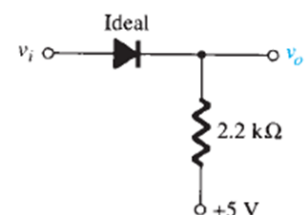
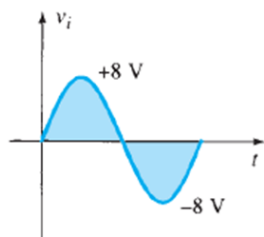


Figure 4

- Q3.** Determine V_0 for each network as shown in Figure 5 and 6 for the input signal. Consider diode is a silicon diode.
[CO2]



Input Signal

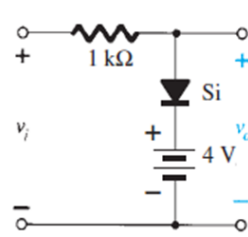


Figure 5

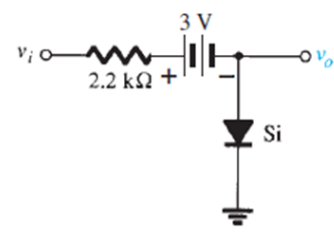
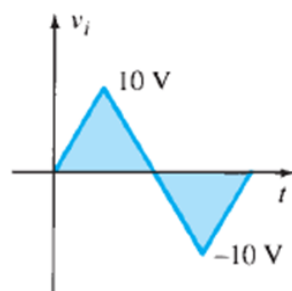


Figure 6

- Q4.** Sketch I_R and V_o for the network is shown in Figure 7 for the input signal. Consider diode is a silicon diode.
[CO2]



Input Signal

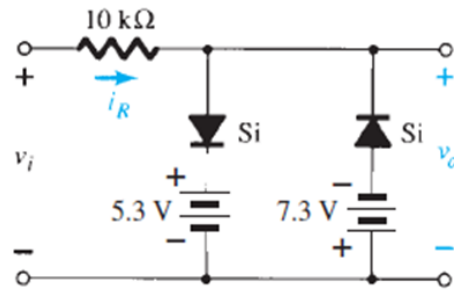
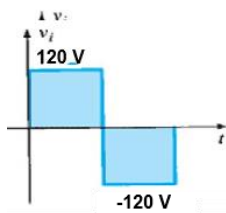


Figure 7

Q5.
[CO2]

Sketch V_o for the network is shown in Figure 8 and 9 for the input signal.



Input Signal

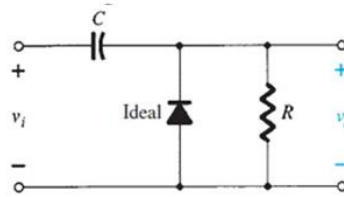


Figure 8

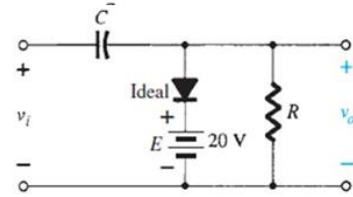
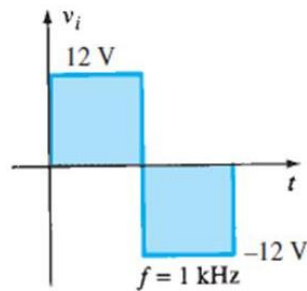


Figure 9

Q6.
[CO2]

For the network of Figure 10.

- Calculate 5 time constant (5τ).
- Compare 5τ to half the period of the applied signal.
- Sketch V_o



Input Signal

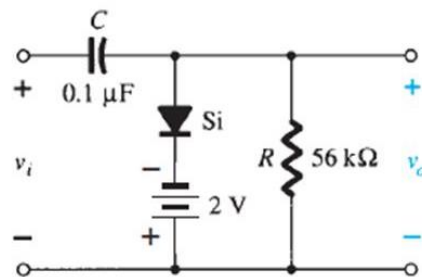


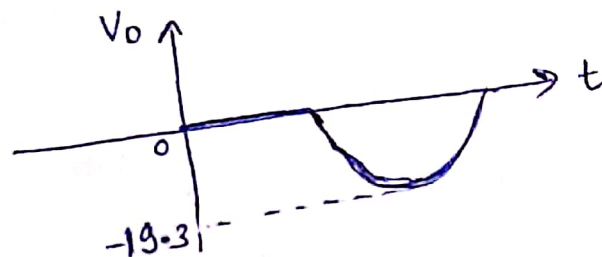
Figure 10

Sol

Q.1

for fig. 1Si diode open for +ve pulse of V_i & $V_o = 0V$ for negative $-20V < V_i \leq -0.7$ diode 'ON' andthe o/p $V_o = V_i + 0.7$ for $V_i = -20V$, $V_o = -20V + 0.7V = -19.3V$ for $V_i = -0.7$, $V_o = -0.7 + 0.7 = 0V$

o/p wave form

for fig. 2if $V_i \leq 8V$, the 8V battery will ensure the diode is F.B. and $V_o = V_i - 8V$ at $V_i = 8V$

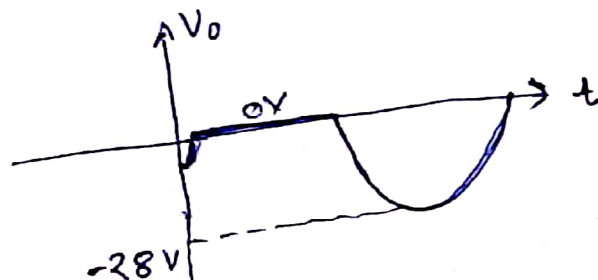
$$V_o = 8V - 8V = 0V$$

at $V_i = -20V$

$$V_o = -20V - 8V = -28V$$

for $V_i > 8V$, diode is R.B. and $V_o = 0$

o/p wave form



Q.2 for fig. 3

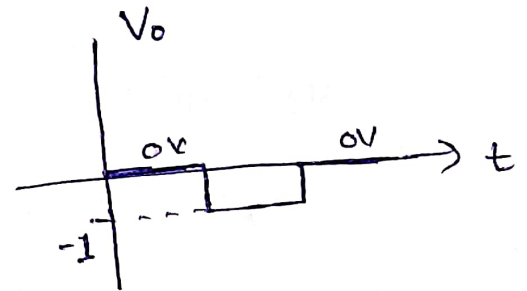
if, $V_i = 20V$, the diode is R.B. and $V_o = 0V$

if, $V_i = -5V$, V_i overpowers the $4V$ battery and the diode is 'ON'

Apply KVL

$$-5V + 4V - V_o = 0$$

$$V_o = -1V$$

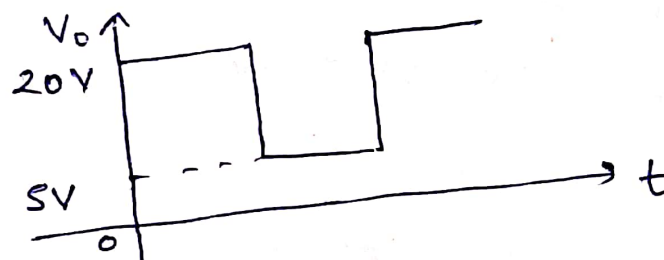


for fig. 4

if, $V_i = 20V$, the $20V$ level overpowers the $5V$ supply and the diode is 'ON'. using the short-circuit equivalent for the diode we find $V_o = V_i = 20V$

if $V_i = -5V$, both V_i and the $5V$ supply R.B. the diode and separate V_i from V_o . However, V_o is connected directly through the $202K$ resistor to the $5V$ supply

o/p
waveform

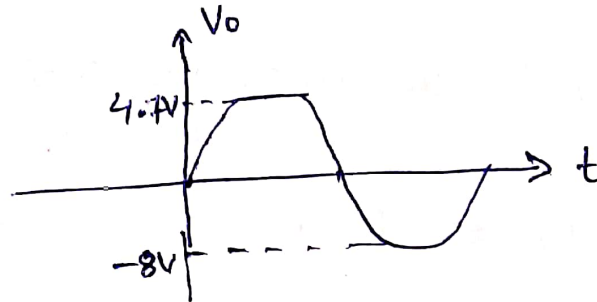


Q.3 For fig. 5

Diode 'ON' if $V_i \geq 4.7V$

for $V_i > 4.7V$, $V_o = 4V + 0.7 = 4.7V$

for $V_i < 4.7V$, diode 'off' & $V_o = V_i$



For fig. 6

Diode 'ON' for $V_i \geq 3.7V$, but V_o now defined as the voltage across the diode

for $V_i \geq 3.7$, $V_o = 0.7V$

for $V_i < 3.7$, diode "off"

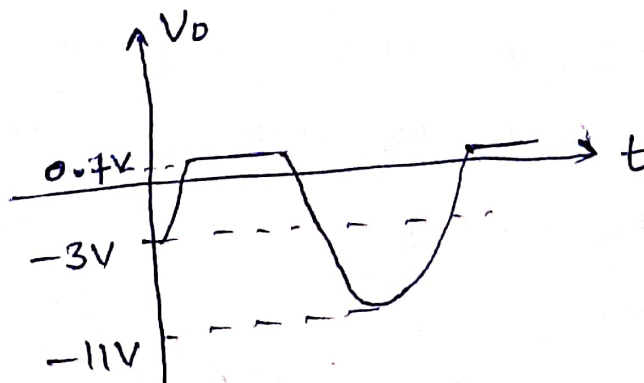
$$I_D = I_R = 0mA \text{ \& } V_{2.2k\Omega} = I_R = 0 \cdot R = 0V$$

therefore, $V_o = V_i - 3V$

at $V_i = 0V$, $V_o = -3V$

& $V_i = -8V$, $V_o = -8 - 3 = -11V$

o/p
waveform



Q.4

for Positive cycle of V_i

The right Si diode is R.B.

The left Si diode is 'ON' for levels of V_i greater than

$$5.3 + 0.7 = 6V, \text{ In fact, } V_o = 6V \text{ for } V_i \geq 6V$$

for $V_i < 6V$ both diodes are R.B. & $V_o = V_i$

for Negative cycle of V_i

The left Si diode is R.B.

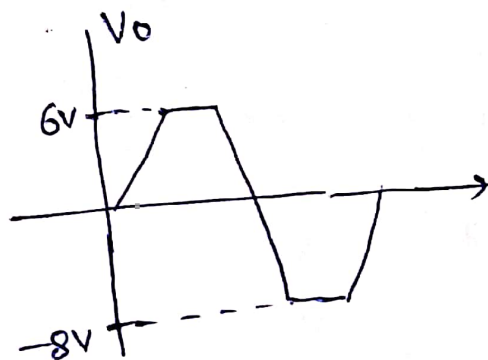
The Right Si diode is 'ON' for levels of V_i

more negative than $7.3 + 0.7 = 8V$.

In fact, $V_o = -8V$ for $V_i \leq -8V$

for $V_i > -8V$ both diodes are R.B. & $V_o = V_i$

q/p
waveform



Analysis for I_R .

for $-8V < V_i < 6V$ there is no conduction through the $10k\Omega$ resistor due to the lack of a complete ckt.

therefore $I_R = 0mA$

for $V_i \geq 6V$

$$V_R = V_i - V_o \Rightarrow V_i - 6V$$

for $V_i = 10V$,

$$V_R = 10 - 6 \Rightarrow 4V$$

$$\& I_R = \frac{4}{10k\Omega} \Rightarrow 0.4mA$$

for $V_i \leq -8V$

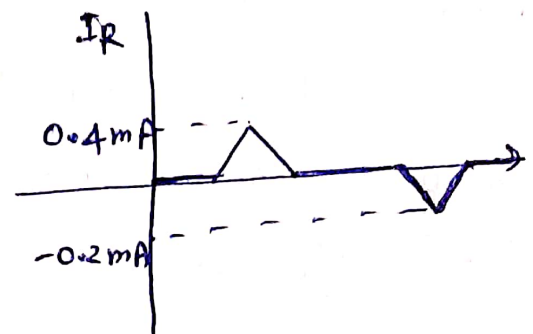
$$V_R = V_i - V_O = V_i + 8V$$

for $V_i = -10V$

$$V_R = -10V + 8V = -2V$$

$$\& I_R = \frac{-2V}{10k\Omega} \Rightarrow -0.2mA$$

waveform



Q.5 for fig. 8

Starting with -ve half cycle.

diode is 'ON' state, capacitor charges to peak value of 120 with polarity $(- \rightarrow +)$. The

output V_o is directly across the 'ON' diode resulting in $V_o = 0$ V

for next +ve half cycle,

$V_o = V_i + 120$ with peak value of

$$V_o = 120 + 120 = 240 \text{ V}$$

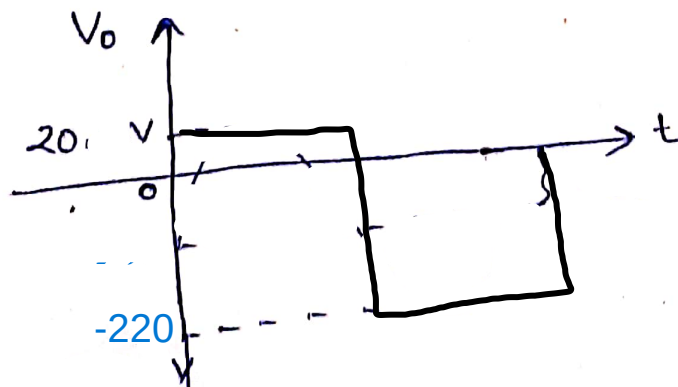
o/p
waveform



for fig. 9 | For +ve half cycle capacitor charges to peak value of $120\text{V} - 20\text{V} = 100$ V with polarity $(+ \rightarrow -)$.

The o/p $V_o = 20\text{V}$.

for next -ve half cycle $V_o = V_i - 100$ V with -ve peak value of $V_o = -120\text{V} - 100 \text{ V} \Rightarrow -220 \text{ V}$



Q.6

(a) $\tau = RC \Rightarrow 58k\Omega \times 0.1 \times 10^{-6} = 5.6 \text{ ms}$

$$5\tau = 5 \times 5.6 \Rightarrow 28 \text{ ms}$$

(b) $5\tau = 28 \text{ ms} \gg \frac{T}{2} = \frac{1 \text{ ms}}{2} = 0.5 \text{ ms}$

$$5\tau : \frac{T}{2} = 56:1$$

(c) for +ve pulse of V_i

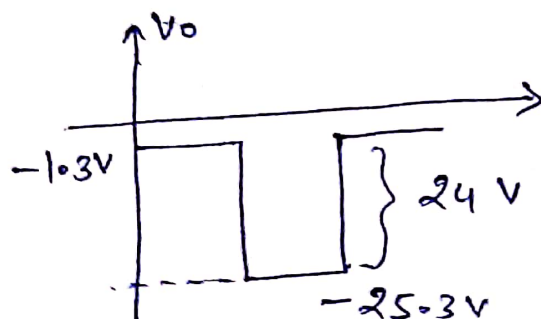
Diode 'ON' & $V_o = -2V + 0.7V = -1.3V$

capacitor charges to $12V + 2V - 0.7V = 13.3$

for -ve pulse of V_i

Diode 'off' and ~~$V_o = 12V + 13.3V = 25.3V$~~

$$V_o = -12V - 13.3V = -25.3V$$



JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
Electronics and Communication Engineering
Basic Electronics (24B11EC111)
Tutorial No. 6

- Q1.** Fig. 1 shows a loaded Zener regulator. If the Zener diode is disconnected, what is the load voltage?
[CO2] [Ans. 16.39 V]

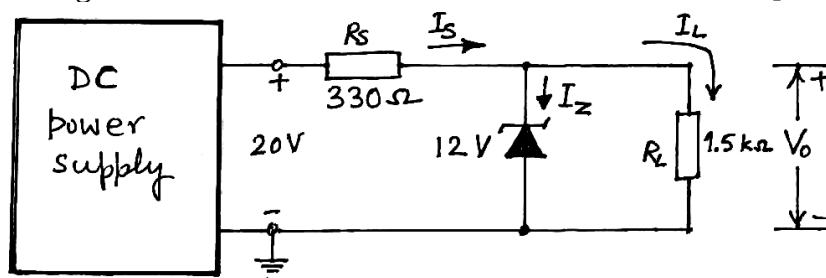


Figure 1

- Q2.** Suppose that the dc supply voltage in the circuit of Fig. 1 slowly decreases from 20 V to 0 V. At some point along the way, the Zener diode will stop regulating. Find the supply voltage where regulation is lost.
[CO2] [Ans. 14.64 V]
- Q3.** Calculate all the three currents in Fig. 1.
[CO2] [Ans. 24.24 mA, 16.24 mA, 8 mA]
- Q4.** Assuming a tolerance of $\pm 10\%$ in both resistors of Fig. 1, what is the maximum Zener current?
[CO2] [Ans. 19.67 mA]
- Q5.** Suppose that the dc supply voltage of Fig. 1 can vary from 20V to 40V. What is the maximum Zener current?
[CO2] [Ans. 76.8 mA]
- Q6.** During the day, the ac supply voltage changes. This causes the output of the unregulated dc power supply to vary from 17.5V to 21V. If the Zener resistance is 11.5Ω , what is the corresponding change in the output voltage of the regulator of Fig. 1?
[CO2] [Ans. 12.1 V to 12.221 V]
- Q7.** What is the minimum load resistance that may be used in Fig. 1 without losing Zener regulation?
[CO2] [Ans. 495 Ω]
- Q8.** The unregulated voltage in Fig. 1 may vary from 16 V to 23 V, and the load current may vary from 1 to 25 mA. Will the Zener regulator stop regulating under these conditions? If so, what is the maximum value of R_S ?
[CO2] [Ans. Yes, 160 Ω]
- Q9.** Figure 2 shows a Zener voltage regulator circuit. Determine V_O , I_L , I_Z and I_S , if $R_L = 180\Omega$.
[CO2] [(Ans. 9 V, 50 mA, 0 mA, 50 mA)]

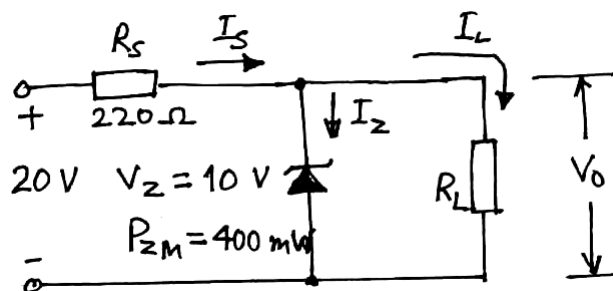


Figure 2

- Q10.** Design the Zener regulator circuit of Fig. 3 to maintain V_L at 12V for a load (I_L) variation from 0 to 200 mA. That is, determine R_S and V_Z . [Ans. 20 Ω , 12V]

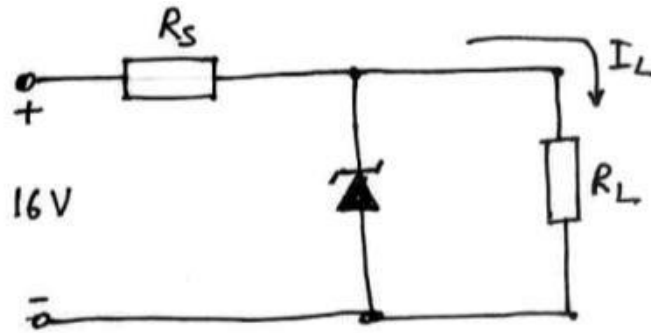


Figure 3

Solutions

Tutorial Sheet - 6

1.

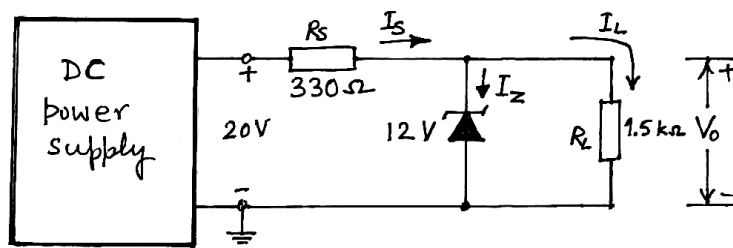


Fig. 1

Applying voltage divider rule, we have

$$V_o = \frac{R_L}{R_S + R_L} V_i = \frac{1500(20)}{1500 + 330} = \mathbf{16.39\text{ V}}$$

2. The regulation stops when the open-circuit (Zener removed) voltage across the load resistor decreases below 12 V. The supply voltage at which this happens is given by

$$V_o = V_i \frac{R_L}{R_L + R_S} = V_Z \quad \Rightarrow \quad V_i = V_Z \frac{R_L + R_S}{R_L} = 12 \times \frac{1500 + 330}{1500} = \mathbf{14.64\text{ V}}$$

3. It is observed from the solution of prob. 1 that $V_o > V_Z$, therefore, Zener diode is in 'on' state and $V_L = V_Z$.

$$I_L = \frac{V_Z}{R_L} = \frac{12\text{ V}}{1.5\text{ k}\Omega} = \mathbf{8\text{ mA}};$$

$$I_S = \frac{V_i - V_Z}{R_S} = \frac{20\text{ V} - 12\text{ V}}{330\text{ }\Omega} = \mathbf{24.24\text{ mA}};$$

$$I_Z = I_S - I_L = 24.24 - 8 = \mathbf{16.24\text{ mA}}$$

4. Zener current I_Z will be maximum when I_L is minimum and I_S is maximum. That is, when R_L is maximum and R_S is minimum.

$$R_{L(\max)} = (1 + 0.1) \times 1.5 \text{ k}\Omega = 1.65 \text{ k}\Omega; \quad R_{S(\min)} = (1 - 0.1) \times 330 \text{ }\Omega = 297 \text{ }\Omega$$

$$I_{L(\min)} = \frac{12 \text{ V}}{1.65 \text{ k}\Omega} = 7.27 \text{ mA}; \quad I_{S(\max)} = \frac{20 - 12}{297} = 26.94 \text{ mA};$$

$$\therefore I_{Z(\max)} = 26.94 - 7.27 = \mathbf{19.67 \text{ mA}}$$

5. Zener current I_Z will be maximum when the dc supply voltage V_i is maximum. For $V_i = 40 \text{ V}$,

$$I_S = \frac{V_i - V_Z}{R_S} = \frac{40 - 12}{330} = 84.8 \text{ mA}; \quad I_L = \frac{12 \text{ V}}{1.5 \text{ k}\Omega} = 8 \text{ mA};$$

$$\therefore I_Z = I_S - I_L = 84.8 - 8 = \mathbf{76.8 \text{ mA}}$$

6.

Since the zener resistance is very small compared to R_S , we can assume that the change in the output voltage is negligibly small while calculating the zener current in the two cases.

For $V_i = 17.5 \text{ V}$:

$$I_S = \frac{17.5 - 12}{330} = 16.67 \text{ mA}; \quad I_L = \frac{12 \text{ V}}{1.5 \text{ k}\Omega} = 8 \text{ mA}; \quad I_Z = I_S - I_L = 8.67 \text{ mA}$$

$$\Delta V_o = I_Z R_Z = (8.67 \text{ mA})(11.5 \text{ }\Omega) = 0.0997 \text{ V} \approx 0.1 \text{ V}$$

$$V_{o1} = V_Z + \Delta V_o = 12 + 0.1 = \mathbf{12.1 \text{ V}}$$

For $V_i = 21 \text{ V}$:

$$I_S = \frac{21 - 12}{330} = 27.27 \text{ mA}; \quad I_L = \frac{12 \text{ V}}{1.5 \text{ k}\Omega} = 8 \text{ mA}; \quad I_Z = I_S - I_L = 19.27 \text{ mA}$$

$$\Delta V_o = I_Z R_Z = (19.27 \text{ mA})(11.5 \text{ }\Omega) = 0.221 \text{ V}$$

$$V_{o1} = V_Z + \Delta V_o = 12 + 0.221 = \mathbf{12.221 \text{ V}}$$

7. The minimum value of load resistance is given as

$$R_{L\min} = \frac{R_S V_Z}{V_i - V_Z} = \frac{330 \times 12}{20 - 12} = \frac{330 \times 12}{8} = \mathbf{495 \text{ }\Omega}$$

8. The Zener diode comes out of breakdown if the voltage across it becomes less than breakdown voltage. For maximum load current and minimum input voltage, the voltage available at the Zener is

$$V_{\min} = V_{i(\min)} - I_{L(\max)} R_S = 16 \text{ V} - (25 \text{ mA})(330) = 8.25 \text{ V}$$

This is less than $V_Z (= 12 \text{ V})$. Hence the circuit stops regulating. For the given conditions, the Zener diode will not come out breakdown if R_S limited to a maximum

$$R_{S\max} = \frac{V_{i(\min)} - V_Z}{I_{L(\max)}} = \frac{16 \text{ V} - 12 \text{ V}}{25 \text{ mA}} = \mathbf{160 \Omega}$$

value.

9. Let us check whether the zener is working in breakdown region. The open-circuit voltage (zener removed) across the load is

$$V = V_i \frac{R_L}{R_L + R_S} = 20 \times \frac{180}{180 + 220} = 9 \text{ V}$$

This is less than V_Z . Hence, the zener is not conducting. $I_Z = \mathbf{0 \text{ mA}}$.

$$I_L = I_S = \frac{20}{220 + 180} = \mathbf{50 \text{ mA}}$$

The output voltage, $V_o = \mathbf{9 \text{ V}}$

10.

Given, $V_L = 12 \text{ V}$, $I_{L\min} = 0$, $I_{L\max} = 200 \text{ mA}$.

The Zener diode must be on to maintain V_L at 12V and V_Z must also be equal to V_L .

Therefore,

$$V_L = V_Z = 12 \text{ V}$$

The minimum value of load resistance to make the Zener diode on is given by

$$R_{L\min} = \frac{R_S V_Z}{V_i - V_Z} = \frac{R_S (12)}{16 - 12} = 3R_S$$

The load current is maximum when load resistance is minimum and given by

$$I_{L\max} = \frac{V_Z}{R_{L\min}}, \quad 200 \text{ mA} = \frac{12}{3R_S},$$

$$R_S = 20\Omega$$

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Tutorial No. 7

Q1. Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$. [Ans. 9, 49, 99]

[CO4]

Q2. Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu A$.

[CO4]

[Ans. 1.02 mA]

Q3. Find the α rating of the transistor shown in Fig. 1. Hence determine the value of I_C using both α and β rating of the transistor.

[CO4]

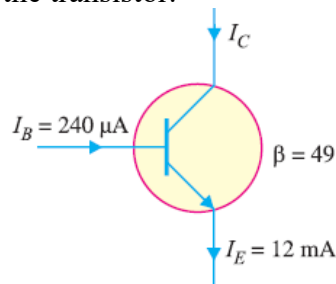


Figure 1

[Ans. 0.98, 11.76 mA]

Q4. In a common base connection, $\alpha = 0.95$. The voltage drops across $2 k\Omega$ resistance which is connected in the collector is 2V. Find the base current.

[CO4]

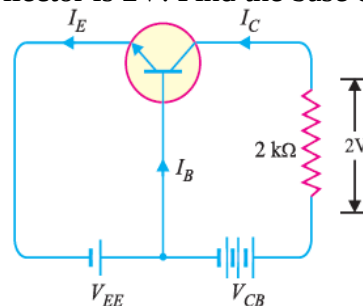


Figure 2

[Ans. 0.05 mA]

Q5. A common base transistor amplifier has an input resistance of 20Ω and output resistance of $100 k\Omega$. The collector load is $1 k\Omega$. If a signal of 500 mV is applied between emitter and base, find the voltage amplification. Assume α_{ac} to be nearly one.

[CO4]

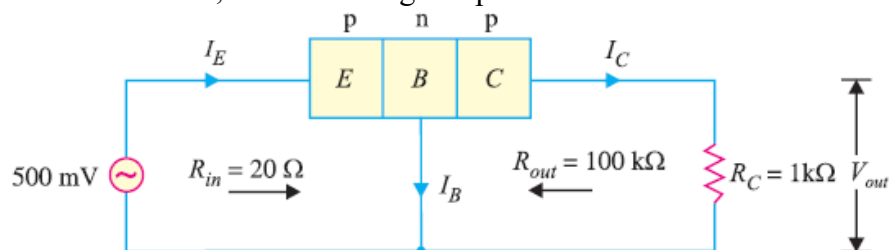


Figure 3

[Ans. 50]

- Q6.** Determine the state of the transistor in Fig. 4 for the following values of collector resistor: (i) $R_C = 2\text{ k}\Omega$ (ii) $R_C = 4\text{ k}\Omega$ (iii) $R_C = 8\text{ k}\Omega$.
[CO4]

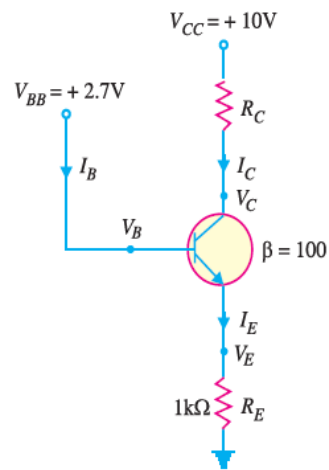


Figure 4

[Ans. Active, Active, Saturation]

Tutorial 7 Solution

Basic Electronics (24B11EC111)

Q1. Find the value of β if (i) $\alpha = 0.9$ (ii) $\alpha = 0.98$ (iii) $\alpha = 0.99$.

Solution :

(i) $\alpha = 0.9$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.9}{1 - 0.9} = 9$$

(ii) $\alpha = 0.98$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.98}{1 - 0.98} = 49$$

(iii) $\alpha = 0.99$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.99}{1 - 0.99} = 99$$

Q2. Calculate I_E in a transistor for which $\beta = 50$ and $I_B = 20 \mu\text{A}$.

Solution :

$$\text{Here } \beta = 50, I_B = 20 \mu\text{A} = 0.02 \text{ mA}$$

$$\text{Now } \beta = \frac{I_C}{I_B}$$

$$\therefore I_C = \beta I_B = 50 \times 0.02 = 1 \text{ mA}$$

$$\text{Using the relation, } I_E = I_B + I_C = 0.02 + 1 = 1.02 \text{ mA}$$

Q3. Find the α rating of the transistor shown in Fig. 1. Hence determine the value of I_C using both α and β rating of the transistor.

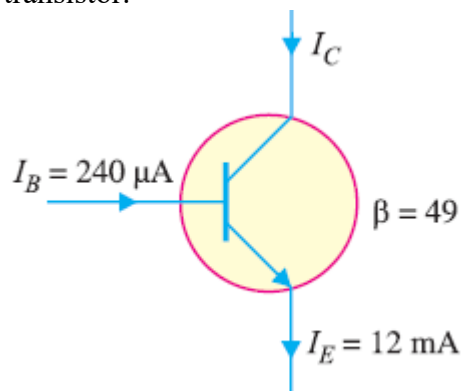


Fig. 1

Solution :

$$\alpha = \frac{\beta}{1+\beta} = \frac{49}{1+49} = \mathbf{0.98}$$

The value of I_C can be found by using either α or β rating as under :

$$I_C = \alpha I_E = 0.98 (12 \text{ mA}) = \mathbf{11.76 \text{ mA}}$$

$$\text{Also } I_C = \beta I_B = 49 (240 \mu\text{A}) = \mathbf{11.76 \text{ mA}}$$

Q4. In a common base connection, $\alpha = 0.95$. The voltage drops across $2 \text{ k}\Omega$ resistance which is connected in the collector is 2V . Find the base current.

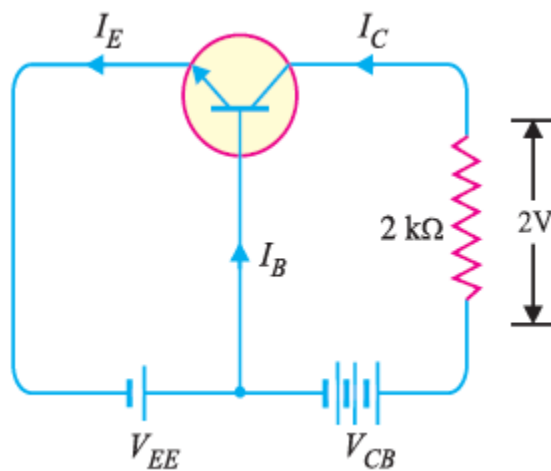


Fig. 2

Solution : Voltage drop across RC ($= 2 \text{ k}\Omega$) is 2V .

$$\therefore I_C = 2 \text{ V} / 2 \text{ k}\Omega = 1 \text{ mA}$$

$$\text{Now } \alpha = I_C / I_E$$

$$\therefore I_E = \frac{I_C}{\alpha} = \frac{1}{0.95} = 1.05 \text{ mA}$$

$$\text{Using the relation, } I_E = I_B + I_C$$

$$\therefore I_B = I_E - I_C = 1.05 - 1 = \mathbf{0.05 \text{ mA}}$$

Q5. A common base transistor amplifier has an input resistance of 20Ω and output resistance of $100 \text{ k}\Omega$. The collector load is $1 \text{ k}\Omega$. If a signal of 500 mV is applied between emitter and base, find the voltage amplification. Assume α_{ac} to be nearly one.

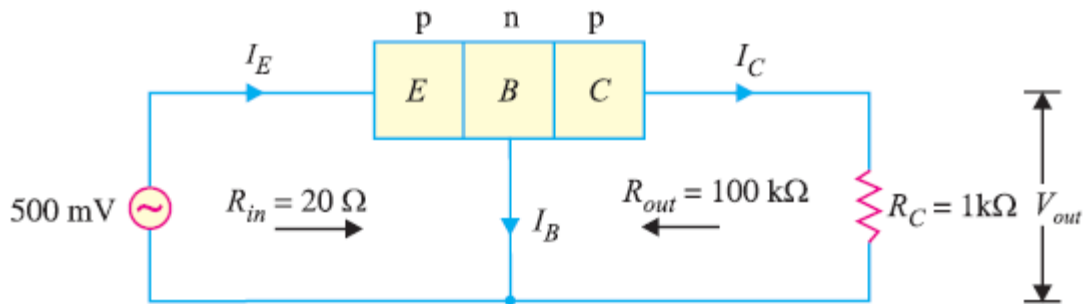


Fig 3

Solution :

Output resistance is very high as compared to input resistance, since the input junction (base to emitter) of the transistor is forward biased while the output junction (base to collector) is reverse biased.

Input current, $I_E = \frac{\text{Signal}}{R_{in}} = \frac{500 \text{ mV}}{20 \Omega} = 25 \text{ mA}$. Since α_{ac} is nearly 1, output current, $I_C = I_E = 25 \text{ mA}$.

Output voltage, $V_{out} = I_C R_C = 25 \text{ mA} \times 1 \text{ k}\Omega = 25 \text{ V}$

$\therefore$ Voltage amplification, $A_v = \frac{V_{out}}{\text{signal}} = \frac{25 \text{ V}}{500 \text{ mV}} = 50$

Q6. Determine the state of the transistor in Fig. 28 for the following values of collector resistor:
(i) $R_C = 2 \text{ k}\Omega$ (ii) $R_C = 4 \text{ k}\Omega$ (iii) $R_C = 8 \text{ k}\Omega$.

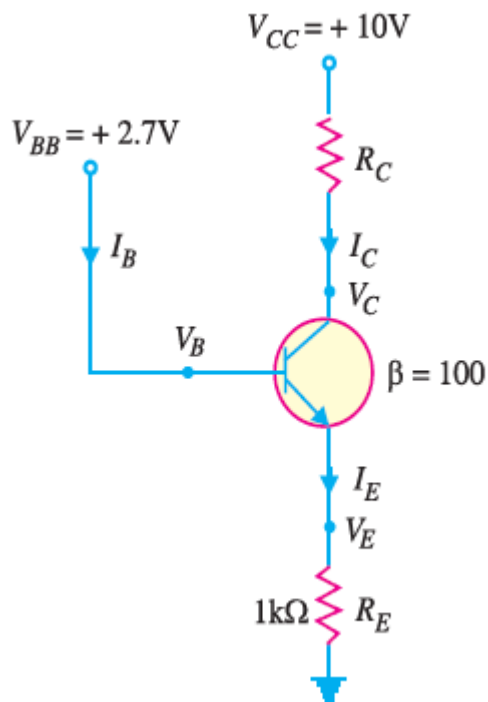


Fig. 4

Solution :

Since I_E does not depend on the value of the collector resistor R_C , the emitter current (I_E) is the same for all three parts.

$$\begin{aligned}\text{Emitter voltage, } V_E &= V_B - V_{BE} = V_{BB} - V_{BE} \\ &= 2.7\text{V} - 0.7\text{V} = 2\text{V}\end{aligned}$$

$$\text{Also } I_E = \frac{V_E}{R_E} = \frac{2\text{V}}{1\text{k}\Omega} = 2\text{ mA}$$

(i) When $R_C = 2\text{ k}\Omega$

Suppose the transistor is active.

$$\therefore I_C = I_E = 2\text{ mA}$$

$$\therefore I_B = I_C / \beta = 2\text{ mA} / 100 = 0.02\text{ mA}$$

$$\begin{aligned}\text{Collector voltage, } V_C &= V_{CC} - I_C R_C \\ &= 10\text{V} - 2\text{ mA} \times 2\text{ k}\Omega = 10\text{V} - 4\text{V} = 6\text{V}\end{aligned}$$

Since $V_C (= 6\text{V})$ is greater than $V_E (= 2\text{V})$, the transistor is active. Therefore, our assumption that transistor is active is correct.

(ii) When $R_C = 4\text{ k}\Omega$

Suppose the transistor is active.

$$\therefore I_C = 2\text{ mA and } I_B = 0.02\text{ mA} \dots \text{ as found above}$$

$$\begin{aligned}\text{Collector voltage, } V_C &= V_{CC} - I_C R_C \\ &= 10\text{V} - 2\text{ mA} \times 4\text{ k}\Omega = 10\text{V} - 8\text{V} = 2\text{V}\end{aligned}$$

Since $V_C = V_E$, the transistor is just at the edge of saturation.

We know that at the edge of saturation, the relation between the transistor currents is the same as in the active state. Both answers are correct.

(iii) When $R_C = 8\text{ k}\Omega$

Suppose the transistor is active.

$$\therefore I_C = 2\text{ mA} ; I_B = 0.02\text{ mA} \dots \text{ as found earlier.}$$

$$\begin{aligned}\text{Collector voltage, } V_C &= V_{CC} - I_C R_C \\ &= 10\text{V} - 2\text{ mA} \times 8\text{ k}\Omega = 10\text{V} - 16\text{V} = -6\text{V}\end{aligned}$$

Since $V_C < V_E$, the transistor is saturated, and our assumption is not correct

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
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Basic Electronics (24B11EC111)
Tutorial No. 8

- Q1.** Consider a process technology for which $L_{\min} = 0.4 \mu\text{m}$, $t_{\text{ox}} = 8 \text{ nm}$, $\mu_n = 450 \text{ cm}^2/\text{V}\cdot\text{s}$, and $V_t = 0.7 \text{ V}$.
[CO4]
- (a) Find C_{ox} and k_n' .
- (b) For a MOSFET with $W/L = 8 \mu\text{m}/0.8 \mu\text{m}$, calculate the values of V_{OV} , V_{GS} , and V_{DSmin} needed to operate the transistor in the saturation region with a dc current $I_D = 100 \mu\text{A}$.
- (c) For the device in (b), find the values of V_{OV} and V_{GS} required to cause the device to operate as a 1000Ω resistor for very small v_{DS} .
- [Ans. $4.32 \times 10^{-3} \text{ F/m}^2$, $194 \mu\text{A/V}^2$, 0.32 V , 1.02 V , 0.32 V , 0.52 V , 1.22 V]**
- Q2.** A circuit designer intending to operate a MOSFET in saturation is considering the effect of changing the device dimensions and operating voltages on the drain current I_D . Specifically, by what factor does I_D change in each of the following cases?
[CO4]
- (a) The channel length is doubled.
- (b) The channel width is doubled.
- (c) The overdrive voltage is doubled.
- (d) The drain-to-source voltage is doubled.
- (e) Changes (a), (b), (c), and (d) are made simultaneously.
- Which of these cases might cause the MOSFET to leave the saturation region?
- [Ans. 0.5; 2; 4; no change; 4; case (c) if v_{DS} is smaller than $2V_{\text{OV}}$]**
- Q3.** Consider an NMOS transistor fabricated in a $0.18\text{-}\mu\text{m}$ process with $L = 0.18 \mu\text{m}$ and $W = 2 \mu\text{m}$. The process technology is specified to have $C_{\text{ox}} = 8.6 \text{ fF}/\mu\text{m}^2$, $\mu_n = 450 \text{ cm}^2/\text{V}\cdot\text{s}$, and $V_{\text{tn}} = 0.5 \text{ V}$.
[CO4]
- (a) Find V_{GS} and V_{DS} that result in the MOSFET operating at the edge of saturation with $I_D = 100 \mu\text{A}$.
- (b) If V_{GS} is kept constant, find V_{DS} that results in $I_D = 50 \mu\text{A}$.
- [Ans. 0.72 V , 0.22 V , 0.06 V]**
- Q4.** Consider NMOS as shown in *Fig. 1*, what will be the change in I_D if V_{DD} changes from 3.3 V to 1.8 V . Consider $V_{\text{IN}} = 2.5 \text{ V}$, $V_{\text{TH}} = 0.5 \text{ V}$ and $\mu_n C_{\text{ox}} W/L = 100 \mu\text{A/V}^2$?
[CO4]

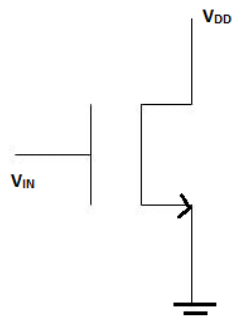


Figure 1

[Ans. 2 μA]

Solution 1

(a)

$$\begin{aligned}C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.45 \times 10^{-11}}{8 \times 10^{-9}} = 4.32 \times 10^{-3} \text{ F/m}^2 \\&= 4.32 \text{ fF}/\mu\text{m}^2 \\k'_n &= \mu_n C_{ox} = 450 \text{ (cm}^2/\text{V} \cdot \text{s)} \times 4.32 \text{ (fF}/\mu\text{m}^2) \\&= 450 \times 10^8 \text{ (}\mu\text{m}^2/\text{V} \cdot \text{s)} \times 4.32 \times 10^{-15} \text{ (F}/\mu\text{m}^2) \\&= 194 \times 10^{-6} \text{ (F/V} \cdot \text{s)} \\&= 194 \mu\text{A/V}^2\end{aligned}$$

(b) For operation in the saturation region,

$$i_D = \frac{1}{2} k'_n \frac{W}{L} V_{ov}^2$$

Thus,

$$100 = \frac{1}{2} \times 194 \times \frac{8}{0.8} V_{ov}^2$$

which results in

$$V_{ov} = 0.32 \text{ V}$$

Thus,

$$V_{GS} = V_t + V_{ov} = 1.02 \text{ V}$$

and

$$V_{DSmin} = V_{ov} = 0.32 \text{ V}$$

(c) For the MOSFET in the triode region with v_{DS} very small,

$$r_{DS} = \frac{1}{k'_n \frac{W}{L} V_{ov}}$$

Thus

$$1000 = \frac{1}{194 \times 10^{-6} \times 10 \times V_{ov}}$$

which yields

$$V_{ov} = 0.52 \text{ V}$$

Thus,

$$V_{GS} = 1.22 \text{ V}$$

Solution 3

First we determine the process transconductance parameter k'_n ,

$$\begin{aligned}k'_n &= \mu_n C_{ox} \\&= 450 \times 10^{-4} \times 8.6 \times 10^{-15} \times 10^{12} \text{ A/V}^2 \\&= 387 \text{ } \mu\text{A/V}^2\end{aligned}$$

and the transistor transconductance parameter k_n ,

$$\begin{aligned}k_n &= k'_n \left(\frac{W}{L} \right) \\&= 387 \left(\frac{2}{0.18} \right) = 4.3 \text{ mA/V}^2\end{aligned}$$

(a) With the transistor operating in saturation,

$$I_D = \frac{1}{2} k_n V_{ov}^2$$

Thus,

$$100 = \frac{1}{2} \times 4.3 \times 10^3 \times V_{ov}^2$$

which results in

$$V_{ov} = 0.22 \text{ V}$$

Thus,

$$V_{GS} = V_m + V_{ov} = 0.5 + 0.22 = 0.72 \text{ V}$$

and since operation is at the edge of saturation,

$$V_{DS} = V_{ov} = 0.22 \text{ V}$$

(b) With V_{GS} kept constant at 0.72 V and I_D reduced from the value obtained at the edge of saturation, the MOSFET will now be operating in the triode region, thus

$$I_D = k_n \left[V_{OV} V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$
$$50 = 4.3 \times 10^3 \left[0.22 V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

which can be rearranged to the form

$$V_{DS}^2 - 0.44 V_{DS} + 0.023 = 0$$

This quadratic equation has two solutions

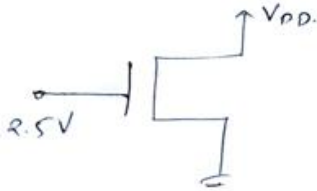
$$V_{DS} = 0.06 \text{ V} \quad \text{and} \quad V_{DS} = 0.39 \text{ V}$$

The second answer is greater than V_{OV} and thus is physically meaningless, since we know that the transistor is operating in the triode region. Thus we have

$$V_{DS} = 0.06 \text{ V}$$

Solution 4

$$V_{th} = 0.5V \quad \mu_n C_{ox} \frac{W}{L} = \frac{100 \mu A}{\sqrt{2}}$$



i) If $V_{DD} = 3.3V$.

$$V_{DS} = 3.3V$$

$$V_{GS} - V_{th} = 2.5 - 0.5 = 2V$$

Saturation Region

$$I_{D1} = \mu_n C_{ox} \frac{W}{L} \left[\frac{V_{GS} - V_{th}}{2} \right]^2$$

$$= \frac{100 \mu A [2.5 - 0.5]^2}{2}$$

$$= 100 \mu A \times 2$$

$$= 200 \mu A$$

ii) If $V_{DD} = 1.8V$.

$$V_{DS} = 1.8V$$

$$V_{GS} - V_{th} = 2.5 - 0.5 = 2V$$

Triode Region

$$I_{D2} = \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2} \right]$$

$$= 100 \mu A \left[2 \times 1.8 - \frac{1.8^2}{2} \right]$$

$$= 100 \mu A [3.6 - 1.62]$$

$$= 198 \mu A$$

Change in $I_D = I_{D1} - I_{D2}$

$$= 200 \mu A - 198 \mu A$$

$$\boxed{\Delta I_D = 2 \mu A}$$

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY
Electronics and Communication Engineering
Basic Electronics (24B11EC111)
Tutorial No. 9

Q1. Assuming the OP-Amp to be ideal, determine the gain of the circuits in Fig. 1(a) and (b)
[CO2]

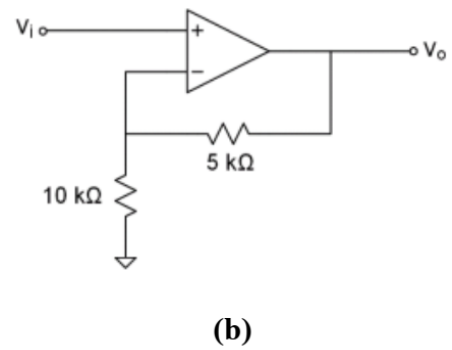
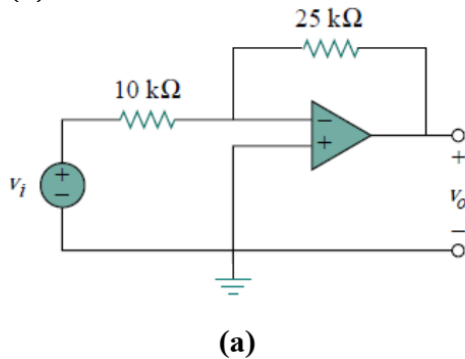


Figure 1

[Ans. -2.5, 1.5]

Q2. Assuming the OP-Amp to be ideal, determine V_a and i in the Fig. 2.
[CO2]

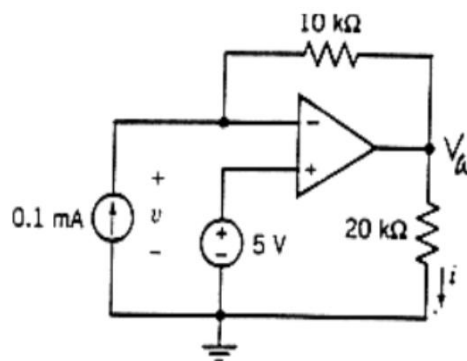


Figure 2

[Ans. 0.2 mA]

Q3. Assuming the OP-Amp to be ideal, determine the gain of the circuit shown in Fig. 3.
[CO2]

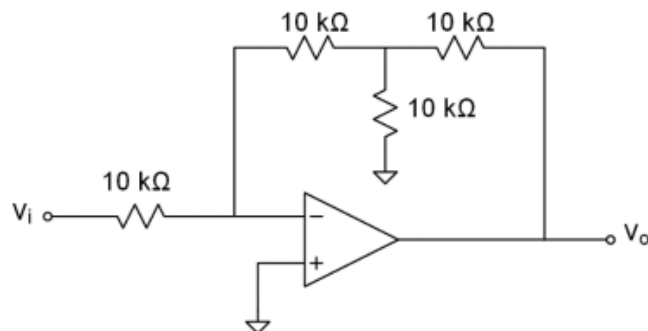


Figure 3

[Ans. -1]

Q4. Assuming the OP-Amp to be ideal, determine the output voltage, V_O in the Fig. 4.
[CO2]

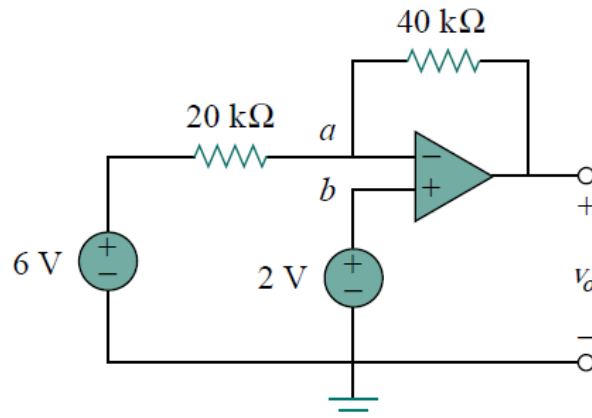


Figure 4

[Ans. -6]

Q5. Assuming the OP-Amp to be ideal, determine i_o and v_o in the Fig. 5
[CO2]

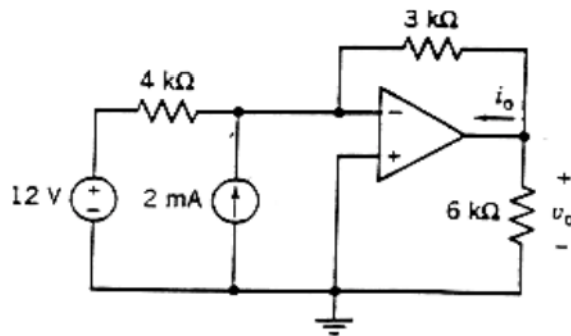


Figure 5

[Ans. 7.5 mA]

Tutorial Sheet 9 solution

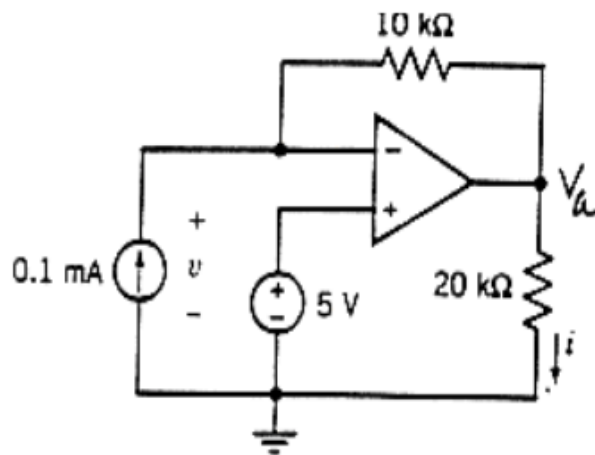
Solution 1(a)

$$\frac{v_o}{v_i} = -\frac{R_f}{R_1} = -\frac{25}{10} = -2.5$$

Solution 1(b)

$$\frac{v_o}{v_i} = \left(1 + \frac{5k}{10k}\right) = \left(1 + \frac{1}{2}\right) = \frac{3}{2}$$

Solution 2



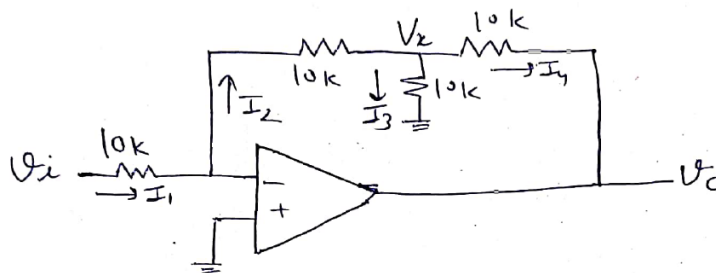
$$v = 5 \text{ V}$$

$$-\left(\frac{v_a - 5}{10000}\right) - 0.1 \times 10^{-3} - 0 = 0$$

$$\Rightarrow v_a = 4 \text{ V}$$

$$i = \frac{v_a}{20000} = \frac{1}{5} \text{ mA}$$

Solution 3



$$I_1 = I_2$$

$$\frac{v_i - 0}{10k} = \frac{0 - v_o}{10k} \Rightarrow v_{in} = -v_o$$

$$I_2 = I_3 + I_4$$

$$I_2 = I_3 + I_4$$

$$\frac{0 - v_x}{10k} = \frac{v_x - 0}{10k} + \frac{v_x - v_o}{10k}$$

$$-v_x = 2v_x - v_o$$

$$v_o = 3v_x = -3v_{in}$$

$$\boxed{\frac{v_o}{v_{in}} = -3}$$

Solution 4

Applying KCL at node a ,

$$\frac{v_a - v_o}{40} = \frac{6 - v_a}{20}$$

$$v_a - v_o = 12 - 2v_a \quad \Rightarrow \quad v_o = 3v_a - 12$$

But $v_a = v_b = 2$ V for an ideal op amp, because of the zero voltage drop across the input terminals of the op amp. Hence,

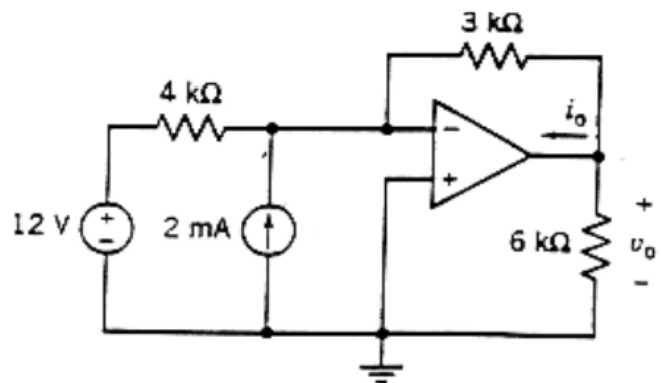
$$v_o = 6 - 12 = -6 \text{ V}$$

Solution 5

$$-\left(\frac{v_o - 0}{3000}\right) - \left(\frac{12 - 0}{4000}\right) - 2 \cdot 10^{-3} = 0$$

$$\Rightarrow v_o = -15 \text{ V}$$

$$i_o + \frac{v_o}{6000} + \frac{v_o}{3000} = 0 \Rightarrow i_o = 7.5 \text{ mA}$$



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Tutorial No. 10

Q1. Assuming the OP-Amp to be ideal, determine v_o and i_o in the Fig. 1.
[CO2]

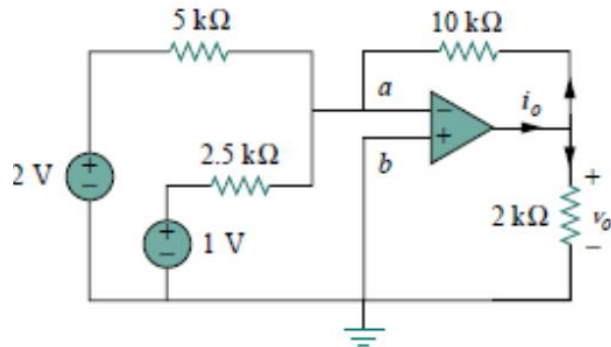


Figure 1

[Ans. -1.2 mA]

Q2. For the difference amplifier circuit shown in Fig. 2, determine the output voltage at terminal A.
[CO2]

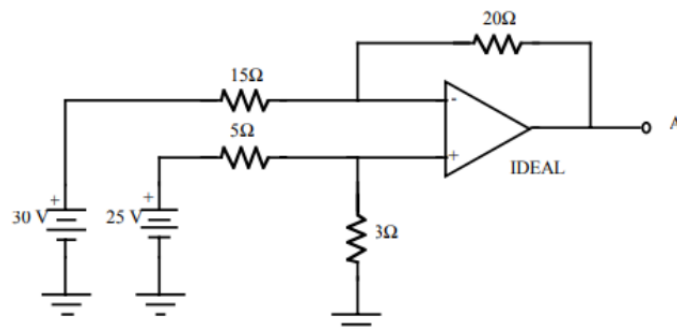


Figure 2

[Ans. -18.125 V]

Q3. Assuming the Op-Amp to be ideal, determine the output voltage, V_o in the Fig. 3.
[CO2]

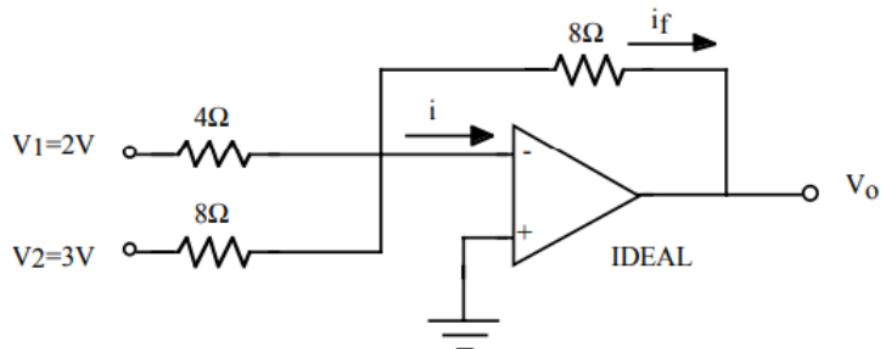


Figure 3

[Ans. -7]

Q4. In the circuit shown in Fig. 4, the Op-AMPs are ideal, determine the output voltage (V_{out}) (in volts)

[CO2]

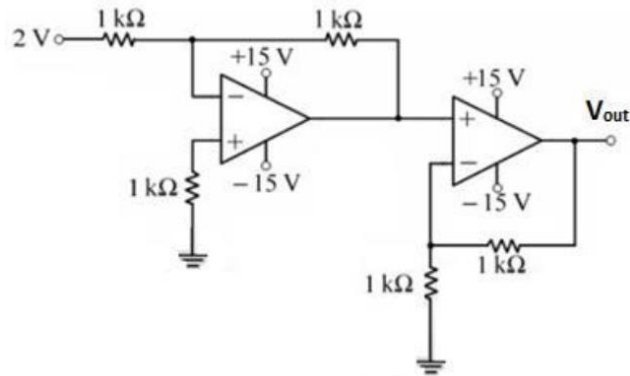


Figure 4

[Ans. -4 V]

Q5. Assuming the OP-Amp to be ideal, the voltage gain of amplifier shown in Fig. 5 (a) and (b)

[CO2]

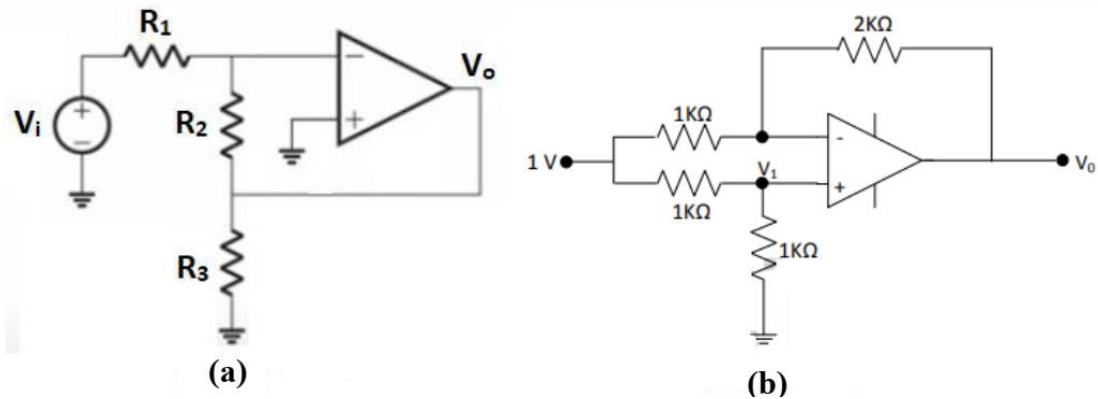
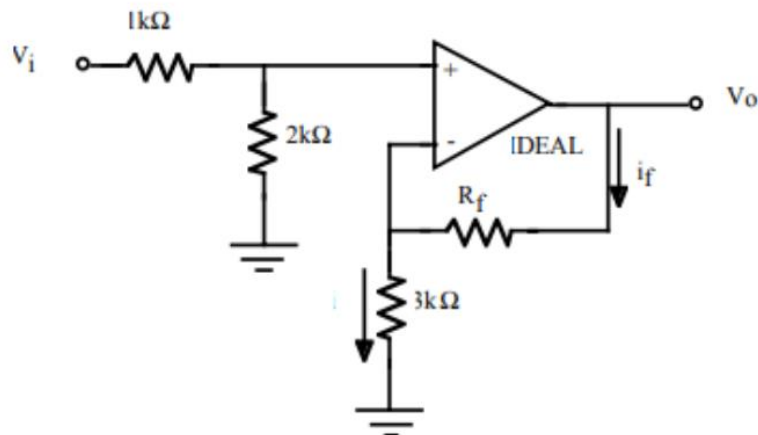


Figure 5

[Ans. $-R_2/R_1$, -0.5 V]

Q6. For the ideal op amp shown in Fig. 6, what should be the value of resistor R_f to obtain a gain of 5?

[CO2]



[Ans. 19.5 k Ω]

Tutorial 10 Solution

Solution 1

This is a summer with two inputs. Using Eq. (5.15),

$$v_o = - \left[\frac{10}{5}(2) + \frac{10}{2.5}(1) \right] = -(4 + 4) = -8 \text{ V}$$

The current i_o is the sum of the currents through the 10-k Ω and 2-k Ω resistors. Both of these resistors have voltage $v_o = -8 \text{ V}$ across them, since $v_a = v_b = 0$. Hence,

$$i_o = \frac{v_o - 0}{10} + \frac{v_o - 0}{2} \text{ mA} = -0.8 - 0.4 = -1.2 \text{ mA}$$

Solution 2

By voltage division,

$$v_{in+} = 25V \left(\frac{3\Omega}{5\Omega + 3\Omega} \right) = 9.375V$$

By the virtual short circuit between the input terminals, $v_{in-} = 9.375 \text{ V}$

Using **Ohm's** law, the current through the 15 Ω resistor is

$$I_{15} = \left(\frac{30V - 9.375V}{15\Omega} \right) = 1.375V$$

The input impedance is infinite; therefore, $I_{in-}=0$ and $I_{15}=I_{20}$.

Use Kirchoff's voltage law to find the output voltage at A.

$$v_A = v_{in-} - 20I_{20} = 9.375 \text{ V} - (20\Omega)(1.375 \text{ A}) = -18.125 \text{ V}$$

Solution 3

This op amp circuit is a summing amplifier. Since $i=0$,

$$i_f = \frac{v_1}{R_1} + \frac{v_2}{R_2} = \frac{3 \text{ V}}{8 \Omega} + \frac{2 \text{ V}}{4 \Omega} = 0.875 \text{ A}$$

$$v_o = -i_f R_f = -(0.875 \text{ A})(8 \Omega) = -7 \text{ V}$$

Solution 4

For OP-AMP -1: It is a inverting Op-Amp.

Therefore,

Apply KCL at inverting terminal

$$(0-V_i)/R_1 + (0-V_{01})/R_2 = 0$$

$$V_{01} = (-R_2/R_1)V_i = -(1k/1k) \times 2 = -2 \text{ Volt} \quad \dots\dots(i)$$

For non-inverting amplifier, apply KCL again at inverting terminal for V_{out}

$$(V_{01}-0)/R + (V_{01}-V_{out})/R_f = 0$$

$$V_{out} = (1+R_f/R) V_{01}$$

$$V_{out} = (1+1/1) \times (-2) = -4 \text{ Volt}$$

Solution 5(a)

For Ideal case of Op-Amp, voltage gain (A_v) = $V_o/V_{id} = \infty$

Therefore, $V_{id} = 0$ i.e. $V_+ - V_- = 0$ or $V_+ = V_-$ (Virtual short concept)

Apply KCL at node A position in the inverting terminal

$$(0-V_i)/R_1 + (0-V_o)/R_2 = 0$$

$$A_v = V_o/V_i = -R_2/R_1$$

Solution 5(b)

Apply superposition in the given Op-Amp

The output voltage (V_{01}) due to inverting input = $(-R_f/R)V_i = (-2k/1K) \times 1 = -2 \text{ Volt}$

and for $V_1 = 1k \times 1/1k+1k = 0.5$ (Potential divider)

The output voltage (V_{02}) due to non inverting input = $(1+R_f/R)V_1 = (1+2/1) \times 0.5 = 1.5 \text{ Volt}$

The total output (V_o) = $V_{01} + V_{02} = -2 + 1.5 = -0.5 \text{ Volt}$

$$A_v = \frac{V_o}{V_i} = \frac{-0.5}{1} = -\frac{1}{2}$$

Solution 6

By voltage division, $v_{in+} = v_i \left(\frac{2k\Omega}{3k\Omega} \right) = \frac{2}{3} v_i$

By the virtual short circuit, $v_{in-} = v_{in+} = \frac{2}{3} v_i$

$$i = \frac{v_{in-}}{3k\Omega} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

Since the op amp draws no current, $i_f = i$

$$\frac{v_o - v_{in-}}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

But, $v_o = 5v_i$.

$$\frac{5v_i - \frac{2}{3} v_i}{R_f} = \frac{\frac{2}{3} v_i}{3k\Omega}$$

$$\frac{13}{3} = \frac{2}{3k\Omega} R_f$$

$R_f = 19.5 \text{ k}\Omega$,